A. JUDUL
Tuliskan judul usulan penelitian maksimal 20 kata

Pengembangan *Refutational-Texts* Berbantuan *Augmented Reality* (RefTaR) untuk Mengurangi Miskonsepsi, *Conceptual Change*, *Cognitive-Psychological Model* , *Demeanor with Physics* Peserta Didik]

B. RINGKASAN
Isian ringkasan penelitian tidak lebih dari 300 kata yang berisi urgensi, tujuan, metode, dan luaran yang ditargetkan

[Dalam pengajaran fisika, diperlukan pemahaman yang kompleks terutama terkait dengan konsep-konsep yang bersifat abstrak karena menjadi dasar untuk pembelajaran yang bermakna. **Urgensi** penelitian ini terletak pada pentingnya mengidentifikasi miskonsepsi, *conceptual change, cognitive-psychological model* siswa dan mahasiswa dalam menghadapi lingkungan belajar yang semakin kompleks. **Penelitian ini bertujuan** mengembangkan dan mengimplementasikan *Refutational-Texts* berbantuan *Augmented Reality (RefTaR)* untuk mengurangi miskonsepsi, *conceptual change*, *cognitive-psychological model*, dan *demeanor with physics* siswa dan mahasiswa. **Penelitian ini menggunakan metode** *Educational Design Research* (EDR) dengan desain ADDIE (*Analysis, Design, Development, Implementation, Evaluation),* sementara desain *embedded experimental model* digunakan dalam implementasinya. Partisipan penelitian ini adalah siswa SMA dan mahasiswa. Tahun pertama difokuskan pada siswa SMA (Banten). Tahun kedua dan ketiga pada mahasiswa penguruan tinggi (Jawa Barat dan Jawa Timur). Analisis data kuantitatif menggunakan analisis Rasch, analisis kualitatif untuk *conceptual change* dan *cognitive-psychological model,* dan analisis NVivo 12 untuk *demeanor with hysics* . Instrumen tes diagnostik yang digunakan adalah *multi-tier* untuk mengidentifikasi level konsepsi siswa dan mahasiswa meliputi: *Sound Understanding* (SU), *Partial Positive* (PP), *Partial Negative* (PN), *Misconception* (MC), *No Understanding* (NU) dan *Un-Code* (UC). Untuk mengidentifikasi *conceptual change* menggunakan kode: *Acceptable Change* (AC), *No Change* (NC), dan *Not Acceptable Change* (NAC). Untuk *Cognitive-psychological Model*digunakan tes level pemahaman secara deskriptif meliputi: *Sound Understanding* (SU), *Partial Understanding* (PU), *Partial Understanding with Alternative Conception* (PU-AC), *Alternative Conception* (AC), dan *No Understanding* (NU). Secara visualisasi meliputi: *Correct Depicting* (CD), *Partial Correct Depicting* (PCD), *Correct Drawing reflecting also non-scientific depicting* (CD-ND), *Incorrect Depicting* (ID), dan *No Depicting* (ND). **Luaran yang ditargetkan** pada tahun pertama sudah Publikasi di **Journal Ilmu Pendidikan Fisika (Sinta 2)**. Tahun kedua:1) Publikasi di (JOTSE) ***Journal of Technology and Science Education* (Q2)**; 2). **Buku Ber-ISBN**. Tahun ketiga publikasi di **Journal of Science Education and Technology** (Q1). Lebih lanjut, **TKT** yang diusulkan mulai dari TKT 1 sampai TKT 3 penelitian RefTaR]

C. KATA KUNCI
Isian 5 kata kunci yang dipisahkan dengan tanda titik koma (;)

[RefTaR; Miskonsepsi; *Conceptual_Change*; Cognitive-Psychological_Mental; Demeanor_*with_Physics*]

D. PENDAHULUAN

Pendahuluan penelitian tidak lebih dari 1000 kata yang terdiri dari:

- *Latar belakang dan rumusan permasalahan yang akan diteliti*
- *Pendekatan pemecahan masalah*
- *State of the art dan kebaruan*
- *Peta jalan (road map) penelitian 5 tahun*

Sitasi disusun dan ditulis berdasarkan sistem nomor sesuai dengan urutan pengutipan.

D.1. LATAR BELAKANG DAN RUMUSAN MASALAH

Tuliskan latar belakang penelitian dan rumusan permasalahan yang akan diteliti, serta urgensi dari dilakukannya penelitian ini

1. [**Latar Belakang**

Kemampuan kognitif memainkan peran yang krusial dalam pembelajaran fisika. Hal tersebut dikarenakan dalam pembelajaran fisika memerlukan pemahaman konsep yang mendalam dan harus sesuai dengan konsepsi ilmiah [1], [2]. Akibatnya, seringkali ditemukan miskonsepsi pada siswa bahkan mahasiswa terkait dengan konsep fisika. Dalam konteks ini, salah satu yang memainkan peran adalah *cognitive-psychological model (model mental)*. Menurut [3], [4], *model mental* dapat dianggap sebagai kerangka kognitif atau struktur pemahaman yang mencakup pengetahuan, keyakinan, dan representasi mental terhadap suatu konsep. Beberapa penelitian sebelumnya menemukan miskonsepsi pada materi momentum impuls [5], [6], listrik [7], gerak [8], suhu dan kalor [9], usaha dan gaya [10], gelombang [11], medan magnet [12] dan perubahan wujud zat [13]. Dengan begitu, penting sekali untuk mengubah miskonsepsi dan *cognitive-psychological model* sesuai dengan konsepsi ilmiah.

Studi pendahuluan yang dilakukan di salah satu SMA Swasta di Tangerang, Banten pada konsep gerak lurus beraturan memperoleh hasil untuk miskonsepsi dengan kategori: *Sound Understanding* (2)%, *Partial Positive (4%)*, *Partial Negative (30), Not Understanding (20%),Misconception (44%*) dan *No Coding* (0%). Sedangkan untuk c*ognitive-psychological model* dengan kategori: Scientific (1%), Synthetic (4%), dan initial (85%). Miskonsepsi (44%) dan initial (85%) merupakan persentase terbesar dan menyatakan bahwa rata-rata peserta didik memiliki konsepsi yang tidak sejalan dengan konsepsi ilmiah. Berdasarkan hasil tersebut, maka penting untuk dilakukan penelitian lanjutan yang mampu mengubah konsepsi dan *cognitive-psychological model* menjadi lebih baik. Salah satu yang paling berperan dalam pembelajaran fisika dan berpotensi untuk digunakan sebagai alternatif pengubah konsepsi dan *cognitive-psychological model* adalah media yang diperkuat melalui pemanfaatan teknologi.

Salah satu pemanfaatan teknologi yang dapat mendukung pembelajaran berupa simulasi dan visualiasi konsep secara menyakinkan adalah aplikasi dari *Augmented Reality* (AR). Aplikasi AR adalah teknologi yang menggabungkan elemen-elemen virtual dengan dunia fisik nyata, menciptakan pengalaman interaktif yang menyatukan dunia nyata dan dunia digital baik berupa 2D ataupun 3D [14], [15]. Implementasi AR memberikan hasil yang positif dalam mendukung pembelajaran[16], [17]. Pemanfaatan elemen AR secara holistik dapat menciptakan lingkungan pendidikan yang secara optimal mendukung *cognitive-psychological model* dan pengubahan konsepsi siswa maupun mahasiswa. Aplikasi AR akan disandingkan dengan *Refutational-Texts* (RT) dalam membantu penguhana konsepsi dan cognitive-psychological model sesuai dengan konsepsi ilmiah. Oleh karena itu, penelitian ini bertujuan mengembangkan dan mengimplementasikan *Refutational-Texts* (RT) berbantuan *Augmented Reality*

(RefTaR) untuk mengubah miskonsepsi, *conceptual change*, *cognitive-psychological model*, *demeanor with physics* peserta didik.

2. Rumusan Permasalahan yang akan Diteliti

Berdasarkan uraian pada latar belakang di atas, adapaun rumusan masalah yang dijabarkan dalam pertanyaan penelitian sebagai berikut:

1. Bagaimana karakteristik *Refutational-Texts* berbantuan *Augmented Reality* (RefTaR) untuk mengubah miskonsepsi, *conceptual change*, *cognitive-psychological model*, *demeanor with physics* peserta didik?
2. Bagaimana *conceptual change*, *cognitive-psychological model*, dan *demeanor with physics* peserta didik setelah diimplementasikan *Refutational-Texts* berbantuan *Augmented Reality* (RefTaR)?
3. Bagaimana efektivitas pembelajaran setelah menerapkan *Refutational-Texts* berbantuan *Augmented Reality* (RefTaR)?

3. Urgensi Penelitian yang dilakukan

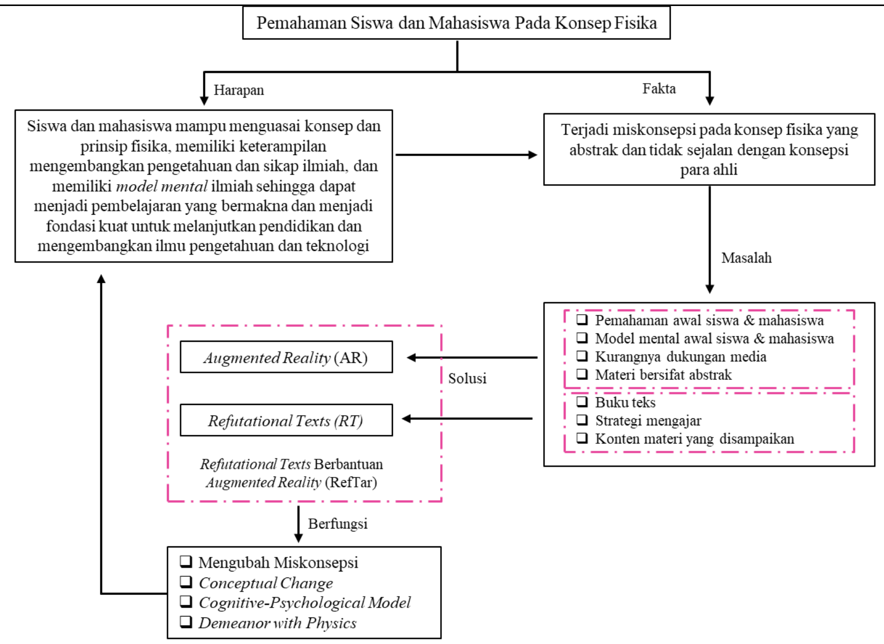
Kemampuan kognitif memainkan peran penting dalam pembelajaran fisika. Hal tersebut dikarenakan pembelajaran fisika tidak hanya melibatkan pengetahuan tentang prinsip-prinsip fisika, akan tetapi juga melibatkan pemahaman konsep. Sesuai dengan hasil penelitian [18]–[20], menyatakan bahwa kemampuan kognitif yang baik sangat berperan dalam memahami prinsip dan konsep yang terdapat dalam pembelajaran fisika. Selain itu, menggabungkan perkembangan kognitif dengan pendidikan fisika membantu peserta didik mengembangkan pemahaman yang relevan dan mendalam tentang konsep-konsep fisika sehingga peserta didik mampu untuk menjelaskan fenomena yang memiliki kaitan dengan fisika [21], [22].

Tambahan lagi, pembelajaran yang bermakna dapat dicapai sesuai dengan teori yang disampaikan oleh Ausubel yaitu ***"Belajar bermakna merupakan suatu proses mengaitkan informasi baru pada konsep-konsep yang terdapat dalam struktur kognitif seseorang"***. Dengan begitu, kemampuan kognitif (*cognitive-psychilogical model*) seperti pemahaman konsep, kemampuan memahami informasi kompleks, dan kemampuan untuk menghubungkan antar konsep menjadi sesuatu yang esensial karena ilmu fisika tidak lepas dari teori, konsep, hukum, dan prinsip fisika[23]–[25].

D.2. PENDEKATAN PEMECAHAN MASALAH

Tuliskan pendekatan dan strategi pemecahan masalah yang telah dirumuskan

Pada dasarnya, siswa maupun mahasiswa seringkali telah memiliki konsepsi awal yang diperoleh dari berbagai sumber seperti lingkungan[26], buku [27], pengalaman belajar [28], dan animasi video [2]. Dalam pembelajaran fisika, terdapat konsep yang bersifat abstrak dan membutuhkan tinjauan secara mikroskopik dan makroskopik. Dengan begitu, solusi yang ditawarkan melalui pendekatan perkembangan teknologi yang mampu memberikan ilustrasi bahkan visualisasi terkait dengan konsep tersebut. Namun, pengunaan AR saja tidak cukup dan membutuhkan panduan yang dapat menhantarkan pada pengubahan konsepsi siswa dan mahasiswa. Dalam hal ini dinamakan *Refutational-Texts* (RT). Pengunaan RT dapat memberikan kesadaran kepada siswa dan mahasiswa, sehingga membantu dalam pengubahan konsepsi dan *cognitive-psychological model* yang tidak sejalan dengan konsepsi ilmiah. Berikut disajikan peta konsep yang dapat menujukkan pendekatan dalam pemecahan masalah pada Gambar 1.

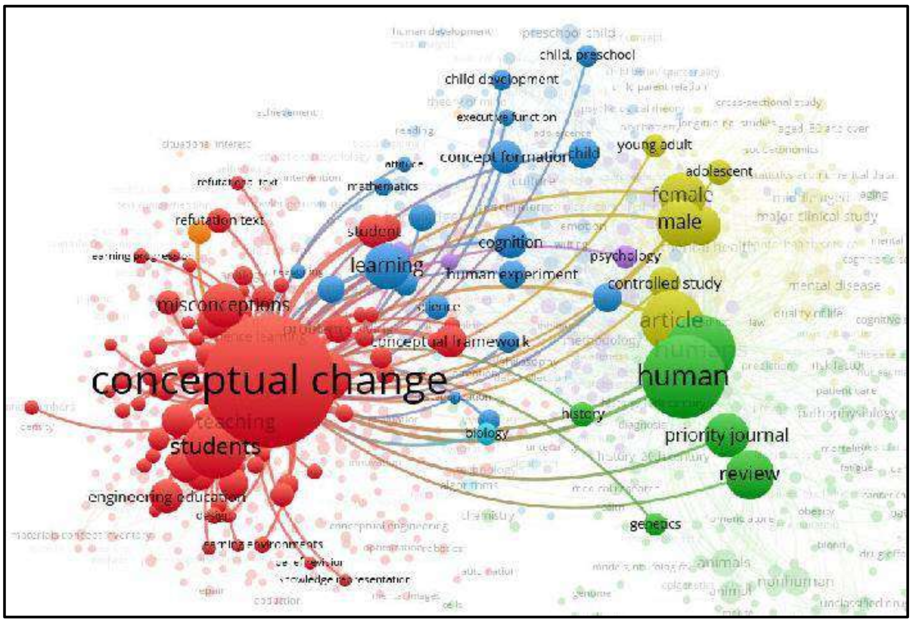


Gambar 1. Pendekatan Pemecahan Masalah RefTar]

D.3. STATE OF THE ART DAN KEBARUAN

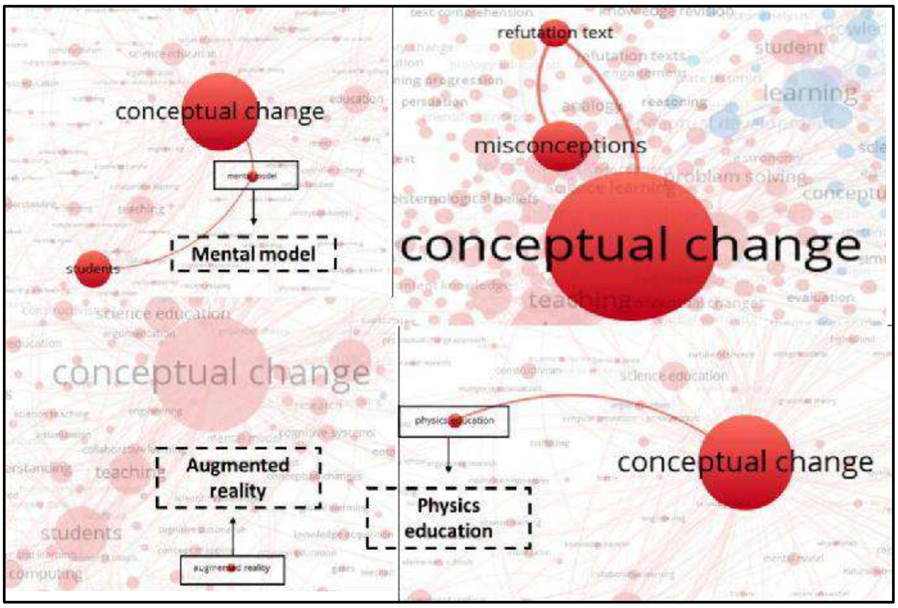
Tuliskan keunggulan dari pemecahan masalah yang ditawarkan pengusul dibandingkan dengan penelitian pengusul sebelumnya atau peneliti lainnya dalam konteks permasalahan yang sama, serta kebaruan usulan dari aspek pendekatan, metode, dsb

[Pendekatan yang digunakan untuk melihat *state-of-the-art* dan kebaruan dalam penelitian ini didasarkan pada analisis *systematic literature review* dan kajian bibliometik terkait penggunaan *Augmented Reality* (AR) dalam *Refutationa- Texts* (RT) pada pembelajaran fisika. Rekomendasi dari penelitian [29], menyatakan ***"In the next study we prefer to use Augmented Reality on refutation text combine".*** Hal tersebut kemudian didukung dengan analisis bibliometrik menggunakan Database Scopus. Melalui hasil pencarian dengan kata kunci "TITLE-ABS-KEY ("*Conceptual Change* " OR "Cognitive-psychilogical model" OR "Refutation text") "All years" didapatkan 4193 dokumen yang ditemukan per tanggal 12 Januari 2024. Hasil tersebut kemudian divisualisasikan menggunakan bantuan *software* VosViewer dan hasilnya disajikan pada Gambar 2.



Gambar 2. Network Visualisasi pada kata *Conceptual Change*

Gambar 2 menyajikan informasi bahwa akurasi kata kunci *conceptual change* ketika difokuskan memiliki hubungan yang lebih sedikit dan terbatas dengan kata kunci yang lain. Misalnya dengan kata kunci *students*, *refutation text*, *misconceptions* dan lain sebagainya. Setelah itu, dilakukan pencarian yang lebih mendalam untuk melihat bagaiamana hubungan dan tren penelitian dan variabel penelitian yang dikaji. Hasil pencarian tersebut dirangkum dan disajikan pada Gambar 3.



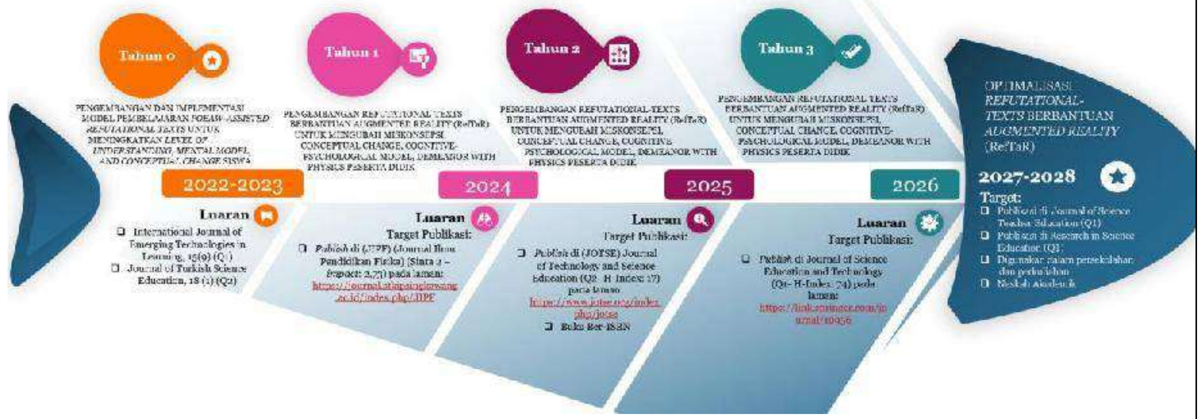
Gambar 3. Visualisasi hubungan variabel penelitian

Berdasarkan Gambar 3, dapat diketahui bagaimana hubungan antar variabel yang menjadi fokus kajian dalam penelitian ini. Pada variabel *modelmental* hubungan erat dengan kata kunci *students* dan *conceptual change*, tetapi memiliki bulatan yang lebih kecil disbanding dengan kedua kata kunci tersebut. Hal ini menandakan kajian pada *model mental* (*cognitive-psychological model*) delmasih belum sebanyak kajian *conceptual change* sehingga menjadi potensi lanjutan untuk diteliti dan dikembangkan lebih lanjut. Selain itu, varibel Cognitive-psychological Modeldan *conceptual change* merupkan dua hal yang berbeda namun masih memiliki hubungan antar lainnya. Selain itu, pada kata kunci *Refutation-Texts* (RT) memiliki hubungan erat juga dengan kata misconception dan *conceptual change*. Hal tersebut sudah tidak asing karena *Refuation-Texts* (RT) memang digunakan untuk mengubah konsepsi dan miskonsepsi [30]–[32]. Disisi lain, kata physics education memiliki hubungan langsung dengan *conceptual change*, tetapi belum memiliki hubungan dengan *model mental* dapat menjadi klaim keterbaruan penelitian ini. Hal tersebut diperkuat dengan kata *Augmented Reality* yang sama sekali tidak memiliki hubungan dengan kata-kata yang lain pada visualisasi yang disajikan. Dengan begitu, **mengintegrasikan *Augmented Reality* pada *Refutational-Texts* (RT) dapat diklaim sebagai keterbaruan pada penelitain yang akan dikaji.**]

D.4. PETA JALAN PENELITIAN

Tuliskan peta jalan penelitian dari tahapan yang telah dicapai, tahapan yang akan dilakukan selama jangka waktu penelitian, dan tahapan yang direncanakan.

Road map dalam penelitian ini mengacu pada peta pengembangan dan penerapan produk *Refutational-Texts* (RT) yang didasarkan pada simulasi atau model pembelajaran yang dilakukan selama lima tahun dan dapat dilihat pada Gambar 4.



Gambar 4. *Roadmap* Penelitian RefTaR]

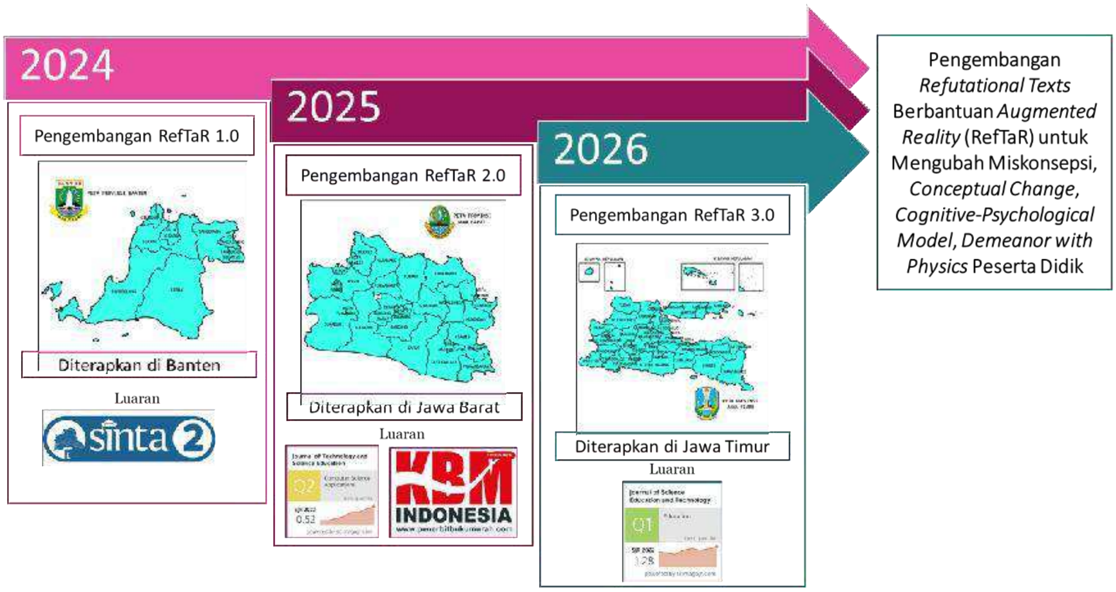
E. METODE

Isian metode atau cara untuk mencapai tujuan yang telah ditetapkan tidak lebih dari 1000 kata. Pada bagian metoda wajib dilengkapi dengan:

- *Diagram alir penelitian yang menggambarkan apa yang sudah dilaksanakan dan yang akan dikerjakan selama waktu yang diusulkan. Format diagram alir dapat berupa file JPG/PNG.*
- *Metode penelitian harus memuat, sekurang-kurangnya proses, luaran, indikator capaian yang ditargetkan, serta anggota tim/mitra yang bertanggung jawab pada setiap tahapan penelitian.*
- *Metode penelitian harus sejalan dengen Rencana Anggaran Biaya (RAB)*

1. [Diagram Alir Penelitian

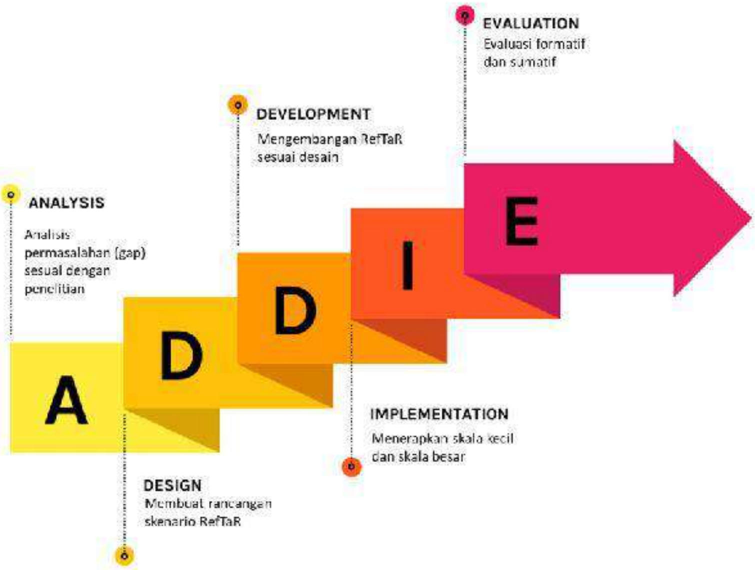
Diagram alir penelitian ini menggambarkan apa yang akan dikerjakan. Secara umun diagram alir penelitian ini ditunjukan pada Gambar 5.



Gambar 5. Diagram Alir Penelitian RefTaR

2. Desain dan Metode Penelitian

Penelitian ini menggunakan metode *Educational Design Research* (EDR) dengan desain ADDIE (*Analysis, Design, Development, Implementation, Evaluation*) dan telah dilakukan oleh beberapa peneliti terkait penelitian pengembangan [33]–[35] seperti pada Gambar 6.



Gambar 6. Desain ADDIE pada Penelitian RefTaR

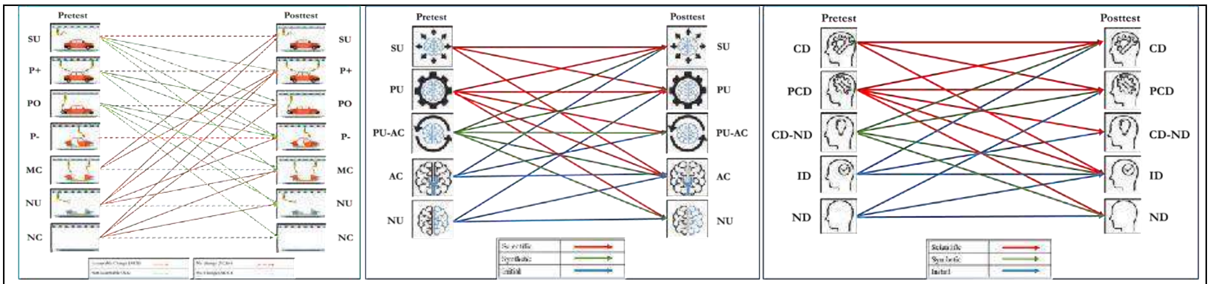
Partisipan dalam penelitian ini terdiri dari siswa tingkat SMA dan mahasiswa Perguruan Tinggi di Indonesia. Penelitian tahun pertama berlokasi di Tangerang, Banten (SMA), tahun kedua di Bandung, Jawa Barat (Perguruan Tinggi), dan tahun ketiga di Surabaya, Jawa Timur (Perguruan Tinggi).

Intrumen tes diagnostik digunakan berjenis *multi-tier* untuk mengidentifikasi level konsepsi siswa meliputi: *Sound Understanding (SU), Partial Positive (PP), Partial Negative (PN), Misconception (MC), No Understanding (NU)* dan *Un-Code (UC) ()*. Untuk mengidentifikasi *Conceptual Change* menggunakan kode: *Acceptable Change (AC), No Change (NC),* dan *Not Acceptable Change (NAC)*[36]. Selanjutnya, untuk *Cognitive-psychological Model*digunakan tes level pemahaman secara desckriptif yang meliputi: *Sound Understanding* (SU), *Partial Understanding* (PU), *Partial Understanding with Alternative Conception* (PU-AC), *Alternative Conception* (AC), dan *No Understanding* (NU). Kemudian secara visualisasi meliputi: *Correct Depicting* (CD), *Partial Correct Depicting* (PCD), *Correct Drawing reflecting also non-scientific depicting* (CD-ND), *Incorrect Depicting* (ID), dan *No Depicting* (ND). Setelah itu dikategorikan mengacu pada rubrik SSI [37] seperti pada Tabel 1.

Tabel1. Evaluasi Model Mental (*cognitive-psychological model)* SSI

Kategori	Kriteria
Scientific	Persepsi yang bertepatan pada level 4 (SU atau CD) atau level 3 (PU atau PCD)
Synthetic	Persepsi yang sebagian bertepatan atau tidak sesuai dengan pengetahuan bertepatan pada level 2 (PU-AC atau CD-ND)
Initial	Persepsi yang tidak sesuai dengan pengetahuan. Jawaban berada pada level 0 (NU atau ND) dan level 1 (AC atau ID)

Analisis data kuantitatif yang direncanakan untuk mengidentifikasi miskonsepsi adalah menggunakan analisis Rasch dan persentase. Sementara, analisis *conceptual change* dan *ognitive-psychological* model dilakukan secara kualitatif dengan beberapa kemungkinan yang dapat terjadi seperti pada Gambar 7.



Gambar 7. Kemungkinan *Conceptual Change* dan *Cognitive-Psychilogical Model*

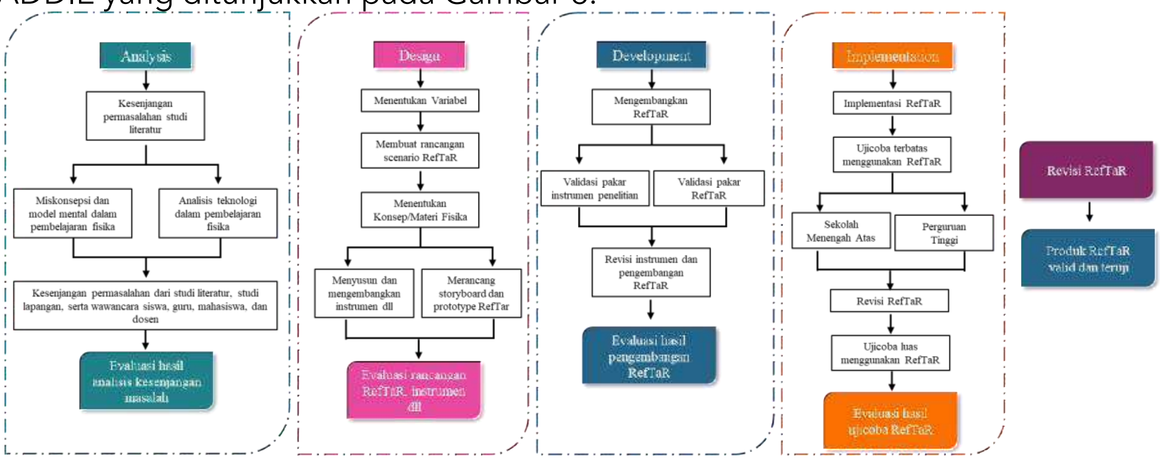
Gambar 7 menunjukkan kemungkinan yang dapat terjadi pada *conceptual change* dan *cognitive-psychological model* siswa dan mahasiswa. Kemudian hasil perubahan konsepsi dan *cognitive-psychological model* berdasarkan kategori tersebut disajikan dengan presertase Perubahan Konsepsi (PK) dan *Cognitive-Psychological Model* (CPM) menggunakan Persamaan 1.

$$PKCPM(\%) = \frac{jumlah\ siswa\ pada\ suatu\ tipe}{jumlah\ seluruh\ siswa} \tag{1}$$

Untuk analisis terkait *demeanor with physics* digunakan analisis NVivo 12 dengan membuat kodifikasi berdasarkan kategori konsepsi siswa dan mahasiswa.

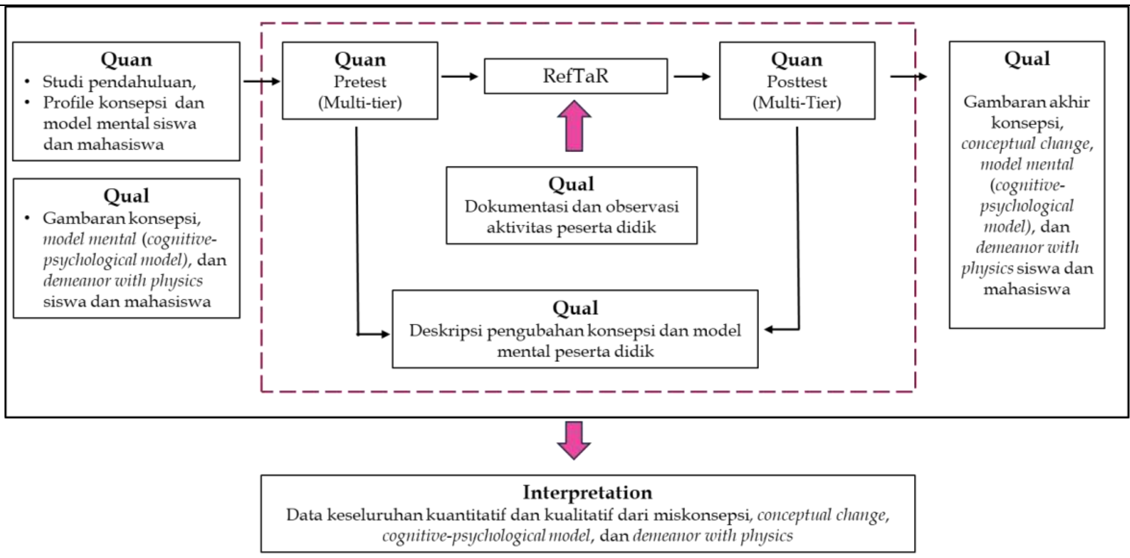
3. Prosedur Penelitian

Prosedur penelitian yang akan dilakukan diturunkan dari desain penelitian ADDIE yang ditunjukkan pada Gambar 8.



Gambar 8. Prosedur Penelitian RefTaR

Tahap *Implementation* merupakan tahap pelaksanaan yang dalam penelitian ini menggunakan *embedded experimental model*. Desain ini mendukung analisis data kuantitatif dengan bantuan data kualitatif [38], [39]. Pada desain ini, penggunaan metode kuantitatif lebih dominan dibandingkan metode kualitatif. Skema desain *embedded experimental model* ditampilkan pada Gambar 9.



Gambar 9. Tahap *Implementation* Menggunakan Desain *Embedded Experimental Model*

Gambar 9 menunjukkan bahwa analisis kualitatif digunakan sebelum dan sesudah perlakuan. Kemudian, masuk pada bagian perlakuan yang dilakukan menggunakan metode quasi eksperimen dengan desain *Nonequivalent Control Group*. Selama perlakuan juga dilakukan analisis kualitatif terkait keterlaknsaan proses perlakuan. Akhirnya sampai pada tahap kuantitatif dilakukan setelah perlakuan. Hasilnya kemudian ddiinterpetasi berdasarkan hasil analisis kuantitatif dan kualitaitf.

4. Hasil Yang Diharapkan

Berdasarkan pemaparan yang telah dijelaskan, maka hasil yang diharapkan adalah menghasilkan produk RefTaR yang valid dan teruji serta dapat menjawab semua rumusan masalah yaitu dapat:1) Mengubah miskonsepsi, *conceptual change, cognitive-psychological model, demeanor with physics* peserta didik; 2) Menghasilkan RefTaR yang valid; 3) Menerapkan RefTaR untuk mengubah *conceptual change, cognitive-psychological model peserta didik*.

5. Indikator Capaian yang Ditargetkan

Sementara, indikator capaian yang ditargetkan dapat dilihat pada Tabel 2.

Tabel 2. Indikator Capaian yang Ditargetkan dalam Penelitian RefTaR

No	Tahun	Jenis	Deskripsi
1.	Pertama	Artikel di Jurnal Nasional (*Published*)	Publikasi di (JIPF) (Journal Ilmu Pendidikan Fisika) (Sinta 2 – *impact*: 2,73) pada laman: https://journal.stkipsingkawang.ac.id/index.php/JIPF 
2.	Kedua	1). Artikel di Jurnal	1). Publikasi di (JOTSE) Journal of Technology and Science Education (Q2- H-Index: 17) pada laman: https://www.jotse.org/index.php/jotse

		Internasional (*accepted/ published*) 2). Buku Ber-ISBN	2). Buku Ber-ISBN 
3.	Ketiga	Artikel di Jurnal Internasional (*accepted/ published*)	Publikasi di Journal of Science Education and Technology (Q1- H-Index: 74) pada laman: https://link.springer.com/journal/10956 

6. Anggota Tim Yang Bertanggung Jawab Pada Setiap Tahapan Penelitian

Anggota tim yang bertanggung jawab pada setiap tahapan penelitian beserta tugasnya dapat dilihat pada Tabel 3.

Tabel 3. Anggota Tim/Mitra yang Bertanggung Jawab pada setiap Tahapan Penelitian

Anggota Tim			
Nama	**NIDN/NIM**	**Bidang Ilmu/Program Studi**	**Peran**
Dr. Achmad Samsudin, M.Pd. (Ketua) Universitas Pendidikan Indonesia	0007108302	Pendidikan Fisika	• Ketua pelaksana (Promotor) • Pelaksana pemantauan utama keberlangsungan penelitian sesuai dengan rancangan dan dinamika yang terjadi selama penelitian 12 jam/minggu. • Pembimbingan arahan dan masukan secara prinsipil dalam setiap tahap penelitian. • Pembimbingan khusus terkait analisis kebutuhan (literatur dan lapangan), pengembangan produk RefTaR hingga uji terbatas dan valid.
Prof. Dr. Andi Suhandi, M.Si. (Anggota) Universitas Pendidikan Indonesia	0017086904	Pendidikan IPA	• Co-Promotor • Membantu ketua pelaksana dalam pemantauan dan monitoring keberlangsungan penelitian10 jam/minggu. • Pembimbingan arahan dan masukan secara prinsipil hingga teknis dalam setiap tahap penelitian.

			• Pembimbingan khusus dalam pengembangan analogi dinamik secara konten fisika dan *Refutational-Texts* Berbantuan *Augmented Reality (RefTaR).*
Mohd Zaidi Bin Amiruddin (Mahasiswa)	2310958	Pendidikan IPA	• Pelaksana utama penelitian 10 jam/minggu. • Survei literatur, studi lapangan, dan menyusun artikel ilmiah. • Mengembangkan instrumen soal, angket dll. • Membuat rancangan RefTaR • Menyusun laporan kemajuan dan laporan akhir.

7. Metode Penelitian Harus Sejalan dengen Rencana Anggmaran Biaya (RAB)

Rencana Anggaran Biaya (RAB) penelitian ini selama tiga tahun. Pada Tahun ke-1 didominasi oleh komponen Pengumpulan Data (27%) dikarenakan merupakan pengembangan awal pengembangan RefTaR 1.0. Pada Tahun ke-2 dan ke-3 didominasi oleh komponen Pelaporan Luaran (31%) karena menuliskan draf artikel hasil dari impelementasi skala terbatas dan skala luas RefTaR 2.0 dan 3.0. Adapun rincian RAB disajikan pada Tabel 4.

Tabel 4. Metode Penelitian Harus Sejalan dengen Rencana Anggaran Biaya (RAB)

Tahun ke-1			
No	**Komponen**	**Total Anggaran**	**%**
1.	Bahan	Rp.10.550.000	18
2.	Pengumpulan Data	Rp.16.000.000	27
3.	Sewa Peralatan	Rp.6.800.000	11
4.	Analisis Data	Rp.12.100.000	20
5.	Pelaporan, Luaran Wajib, dan Luaran Tambahan	Rp.14.550.000	24
Total		Rp.60.000.000	100
Tahun ke-2			
No	**Komponen**	**Total Anggaran**	**%**
1	Bahan	Rp.10.500.000	18
2	Pengumpulan Data	Rp.9.950.000	17
3	Sewa Peralatan	Rp.6.800.000	11
4	Analisis Data	Rp.14.000.000	23
5	Pelaporan, Luaran Wajib, dan Luaran Tambahan	Rp.18.750.000	31
Total		Rp.60.000.000	100
Tahun ke-3			
No	**Komponen**	**Total Anggaran**	**%**
1	Bahan	Rp.7.550.000	13
2	Pengumpulan Data	Rp.15.000.000	25
3	Sewa Peralatan	Rp.6.800.000	11
4	Analisis Data	Rp.12.300.000	21

5	Pelaporan, Luaran Wajib, dan Luaran Tambahan	Rp.18.800.000	31
Total		Rp.60.000.000	100

F. JADWAL PENELITIAN

Jadwal penelitian disusun berdasarkan pelaksanaan penelitian, harap disesuaikan berdasarkan lama tahun pelaksanaan penelitian

Tahun ke-1

No	Nama Kegiatan	Bulan											
		1	2	3	4	5	6	7	8	9	10	11	12
1	**Persiapan Penelitian**												
	Survei literatur dan analisis peluang penggunaan *Refutational-Texts* Berbantuan *Augmented Reality* (RefTaR)												
	FGD penyusunan proposal RefTaR Tahun ke-1												
	Penyelarasan, finalisasi, dan administrasi substansi proposal RefTaR Tahun ke-1												
	Penyusunan instrumen tes diagnostik *multi-tier* dan level pemahaman RefTaR 1.0												
	Diagnosis miskonsepsi dan Cognitive-psychological Modelsiswa												
2	**Pengembangan RefTaR 1.0**												
	Pembuatan desain dan *story board* RefTaR 1.0												
	Pembuatan perangkat pembelajaran (RPP, skenario pembelajaran, LKPD, dan bahan ajar lainya) RefTaR 1.0												
	Validasi hasil pengembangan RefTaR 1.0 dan perangkat pembelajaran												
3	**Pelaksanaan Penelitian RefTaR 1.0**												
	Implementasi pada tingkatan SMA di Tangerang, Banten Indonesia												
	Pengelolahan data berdasarkan hasil *multi-tier-test*												
	Analisis data dan pembahasan berdasarkan *multi-tier-test* RefTaR 1.0												
4	**Finalisasi Hasil**												
	Penyusunan laporan kemajuan												
	Seminar progress penelitian												
	Penyusunan laporan akhir												
	Menulis artikel ilmiah												
	Publikasi di (JIPF) (Journal Ilmu Pendidikan Fisika) (Sinta 2 – impact: 2,73)												

Tahun ke-2

No	Nama Kegiatan	Bulan											
		1	2	3	4	5	6	7	8	9	10	11	12
1	**Persiapan Penelitian**												
	FGD penyusunan proposal RefTaR Tahun ke-2												
	Penyelarasan, finalisasi, dan administrasi substansi proposal RefTaR Tahun ke-2												
	Penyusunan instrumen tes diagnostik *multi-tier* dan level pemahaman RefTaR 2.0												
	Diagnosis miskonsepsi dan *cognitive-psychological model* pada mahasiswa												
2	**Pengembangan RefTaR 2.0**												
	Pengembangan desain dan *story board* RefTaR 2.0												
	Pengembangan model pembelajaran RefTaR 2.0												
	Pembuatan perangkat pembelajaran (RPS, skenario pembelajaran, LKM, dan bahan ajar lainya) RefTaR 2.0												
	Validasi hasil pengembangan RefTaR 2.0 dan perangkat pembelajaran												
3	**Pelaksanaan Penelitian RefTaR 2.0**												
	Implementasi pada tingkatan Perguruan Tinggi di Kota Bandung, Indonesia												
	Pengelolahan data berdasarkan hasil *multi-tier-test* pada RefTaR 2.0												
	Analisis data dan pembahasan berdasarkan *multi-tier-test* RefTaR 2.0												
4	**Finalisasi Hasil**												
	Penyusunan laporan kemajuan												
	Seminar progress penelitian												
	Penyusunan laporan akhir												
	Menulis artikel ilmiah												
	Publikasi di (JOTSE) Journal of Technology and Science Education (Q2- H-Index: 17)												
	Terbit Buku Ber-ISBN												

Tahun ke-3

No	Nama Kegiatan	Bulan											
		1	2	3	4	5	6	7	8	9	10	11	12
1	**Persiapan Penelitian**												
	FGD penyusunan proposal RefTaR Tahun ke-3												
	Penyelarasan, finalisasi, dan administrasi substansi proposal RefTaR Tahun ke-3												

	Penyusunan instrumen diagnostik *multi-tier-test* dan level pemahaman RefTaR 3.0												
	Diagnosis miskonsepsi dan *cognitive-psychological model* pada mahasiswa												
2	**Pengembangan RefTaR 3.0**												
	Pengembangan desain dan *story board* RefTaR 3.0												
	Perbaikan model pembelajaran RefTaR 2.0 menjadi RefTaR 3.0												
	Perbaikan perangkat pembelajaran (RPS, skenario pembelajaran, LKM, dan bahan ajar lainya) RefTaR 3.0												
3	**Pelaksanaan Penelitian RefTaR 3.0**												
	Implementasi pada tingkatan Perguruan Tinggi di Kota Surabaya, Indonesia												
	Pengelolahan data berdasarkan hasil *multi-tier-test* pada RefTaR 3.0												
	Analisis data dan pembahasan berdasarkan *multi-tier-test* RefTaR 3.0												
4	**Finalisasi Hasil**												
	Penyusunan laporan kemajuan												
	Seminar progress penelitian												
	Penyusunan laporan akhir												
	Menulis artikel ilmiah												
	Publikasi di Journal of Science Education and Technology (Q1- H-Index: 74)												

G. DAFTAR PUSTAKA

Sitasi disusun dan ditulis berdasarkan sistem nomor sesuai dengan urutan pengutipan. Hanya pustaka yang disitasi pada usulan penelitian yang dicantumkan dalam Daftar Pustaka.

[[1] M. T. H. Chi, J. D. Slotta, and N. De Leeuw, "From things to processes: A theory of conceptual change for learning science concepts," *Learn. Instr.*, vol. 4, no. 1, pp. 27–43, 1994.

[2] C. Kulgemeyer and J. Wittwer, "Misconceptions in physics explainer videos and the illusion of understanding: An experimental study," *Int. J. Sci. Math. Educ.*, vol. 21, no. 2, pp. 417–437, 2023.

[3] S. A. Gray, E. Zanre, and S. R. J. Gray, "Fuzzy cognitive maps as representations of mental models and group beliefs," in *Fuzzy cognitive maps for applied sciences and engineering: From fundamentals to extensions and learning algorithms*, Springer, 2013, pp. 29–48.

[4] P. N. Johnson-Laird, "Mental models and cognitive change," *J. Cogn. Psychol.*, vol. 25, no. 2, pp. 131–138, 2013.

[5] W. E. Ardhanariswari, W. N. Wulandari, H. I. Khalidah, and B. Setiaji, "Analysis of High School Students' Learning Difficulties in Understanding the Mechanics Concept," *Impuls. J. Res. Innov. Phys. Educ.*, vol. 1, no. 2, pp. 72–79, 2021.

[6] R. Adimayuda, A. H. Aminudin, I. Kaniawati, E. Suhendi, and **A. Samsudin**, “A multitier open-ended momentum and impulse (MOMI) instrument: Developing and assessing quality of conception of 11th grade sundanese students with rasch analysis,” *Int. J. Sci. Technol. Res.*, vol. 9, no. 2, pp. 4799–4804, 2020.
[7] J. Addido, A. Burrows, and T. Slater, “The Effect of the Conceptual Change Model on Conceptual Understanding of Electrostatics,” *Educ. Sci.*, vol. 12, no. 10, p. 696, 2022.
[8] N. Nurhasanah, **A. Samsudin**, S. Linuwih, and A. D. Sutrisno, “Is SiPOERT an Innovative Learning? Implementation and Identification of Students’ Conception on Simple Harmonic Motion,” *Indones. J. Learn. Adv. Educ.*, vol. 5, no. 1, pp. 78–91, 2022.
[9] H. E. Haryono, K. N. Aini, **A. Samsudin**, and P. Siahaan, “Using a three-level diagnostic test instrument to detect student misconception: A case of the concept of heat,” in *AIP Conference Proceedings*, AIP Publishing, 2023.
[10] **A. Samsudin** *et al.*, “Reconstructing students’ misconceptions on work and energy through the PDEODE* E tasks with think-pair-share,” *J. Turkish Sci. Educ.*, vol. 18, no. 1, pp. 118–144, 2021.
[11] S. N. Mufida, **A. Samsudin**, E. Suhendi, I. Kaniawati, and H. Novia, “CoSiReT: Innovation of ReT (Refutation Texts) to Reduce Students’ Misconceptions Concerning Transverse Waves,” *J. Penelit. Pendidik. IPA*, vol. 9, no. 11, pp. 9363–9371, 2023.
[12] **A. Samsudin**, A. Suhandi, D. Rusdiana, I. Kaniawati, and B. Coştu, “Investigating the effectiveness of an active learning based-interactive conceptual instruction (ALBICI) on electric field concept,” in *Asia-Pacific Forum on Science Learning and Teaching*, The Education University of Hong Kong, Department of Science and …, 2016, pp. 1–41.
[13] F. Zheng and Y. Zhang, “Pre-concept Inquiry of" Matter State Changes" in Junior High School Science,” *Asian J. Educ. Soc. Stud.*, vol. 48, no. 4, pp. 73–80, 2023.
[14] A. Rueda Márquez de la Plata, P. A. Cruz Franco, and J. A. Ramos Sánchez, “Applications of Virtual and Augmented Reality Technology to Teaching and Research in Construction and Its Graphic Expression,” *Sustainability*, vol. 15, no. 12, p. 9628, 2023.
[15] S. Dargan, S. Bansal, M. Kumar, A. Mittal, and K. Kumar, “Augmented reality: A comprehensive review,” *Arch. Comput. Methods Eng.*, vol. 30, no. 2, pp. 1057–1080, 2023.
[16] P. Yaniawati, R. Indrawan, and M. P. Mubarika, “The Potential of Mobile Augmented Reality as a Didactic and Pedagogical Source in Learning Geometry 3D.,” *J. Technol. Sci. Educ.*, vol. 13, no. 1, pp. 4–22, 2023.
[17] T. Koparan, H. Dinar, E. T. Koparan, and Z. S. Haldan, “Integrating augmented reality into mathematics teaching and learning and examining its effectiveness,” *Think. Ski. Creat.*, vol. 47, p. 101245, 2023.
[18] E. F. Redish, “Implications of cognitive studies for teaching physics,” *Am. J. Phys.*, vol. 62, no. 9, pp. 796–803, 1994.
[19] M. I. Simeon, M. A. Samsudin, and N. Yakob, “Effect of design thinking approach on students’ achievement in some selected physics concepts in the context of STEM learning,” *Int. J. Technol. Des. Educ.*, pp. 1–28, 2020.
[20] F. Mutohhari, S. Sutiman, M. Nurtanto, N. Kholifah, and A. Samsudin, “Difficulties in Implementing 21st Century Skills Competence in Vocational Education Learning.,” *Int. J. Eval. Res. Educ.*, vol. 10, no. 4, pp. 1229–1236,

2021.
[21] O. M. Ogundeji, B. C. Madu, and C. C. Onuya, "Scientific Explanation of Phenomena and Concept Formation as Correlates of Students' Understanding of Physics Concepts.," *Eur. J. Phys. Educ.*, vol. 10, no. 3, pp. 10–19, 2019.
[22] A. J. Richards, D. C. Jones, and E. Etkina, "How students combine resources to make conceptual breakthroughs," *Res. Sci. Educ.*, vol. 50, pp. 1119–1141, 2020.
[23] D. Panagou, K. T. Kotsis, and G. Stylos, "An empirical study on the evolution of students' perceptions in basic concepts of physics of primary and secondary education in Cyprus," *Electron. J. Res. Sci. Math. Educ.*, vol. 26, no. 2, pp. 91–109, 2022.
[24] N. E. Susilowati, M. Muslim, R. Efendi, and **A. Samsudin**, "PISA 2021 creative thinking instrument for students: Physics teachers' perceptions," *Indones. J. Sci. Math. Educ.*, vol. 5, no. 2, pp. 194–209, 2022.
[25] R. Puspitasari and F. Mufit, "Conditions of learning physics and students' understanding of the concept of motion during the covid-19 pandemic," in *Journal of Physics: Conference Series*, IOP Publishing, 2021, p. 12045.
[26] J. Siantuba, L. Nkhata, and T. de Jong, "The impact of an online inquiry-based learning environment addressing misconceptions on students' performance," *Smart Learn. Environ.*, vol. 10, no. 1, p. 22, 2023.
[27] E. Taslidere and B. Yıldırım, "Effect of conceptual change-oriented instruction on students' conceptual understanding and attitudes towards simple electricity," *Int. J. Sci. Math. Educ.*, vol. 21, no. 5, pp. 1567–1589, 2023.
[28] A. H. Putri, A. Suhandi, and **A. Samsudin**, "Facilitating three levels of representation through visual media: a pilot study on constructivism learning," *J. Southwest Jiaotong Univ.*, vol. 58, no. 4, 2023.
[29] M. Sakulphon, N. Srisawasdi, and P. Wangsomnuk, "Examining relationship between biology attitudes and perceptions toward mobile augmented reality of photosynthesis and impact on gender difference," in *Proceedings of the 23rd International Conference on Computers in Education*, Asia-Pacific Society for Computers in Education China, 2015, pp. 836–841.
[30] C. D. Tippett, "Refutation text in science education: A review of two decades of research," *Int. J. Sci. Math. Educ.*, vol. 8, pp. 951–970, 2010.
[31] G. M. Aguilar, S. J., Polikoff, M. S., & Sinatra, "Refutation Texts: A New Approach to Changing Public Misconceptions About Education Policy," no. July, 2019, doi: 10.3102/0013189X19849416.
[32] D. H. Palmer, "Investigating the relationship between refutational text and conceptual change," *Sci. Educ.*, vol. 87, no. 5, pp. 663–684, 2003.
[33] M. R. Idris and R. Che'Rus, "Developing the Rekatronik Module Using ADDIE Model for Electronic Design Topic in the Design and Technology (D&T) Subject," *Asian J. Res. Educ. Soc. Sci.*, vol. 5, no. 2, pp. 113–122, 2023.
[34] Ü. Kartoğlu, R. C. Siagian, and T. C. Reeves, "Creating a 'Good clinical practices inspection' authentic online learning environment through educational design research," *TechTrends*, vol. 64, no. 4, pp. 616–627, 2020.
[35] S. McKenney and T. Reeves, *Conducting educational design research*. Routledge, 2018.
[36] A. H. Aminudin, İ. Kaniawati, E. Suhendi, **A. Samsudin**, B. Coştu, And R. Adimayuda, "Rasch analysis of Multitier Open-ended Light-Wave

Instrument (MOLWI): Developing and assessing second-years sundanese-scholars alternative conceptions," *J. Educ. Gift. Young Sci.*, vol. 7, no. 3, pp. 557–579, 2019.
[37] M. A. Kurnaz and C. Eksi, "An analysis of high school students' mental models of solid friction in physics," *Educ. Sci. Theory Pract.*, vol. 15, no. 3, 2015.
[38] J. W. Creswell and J. D. Creswell, *Research design: Qualitative, quantitative, and mixed methods approaches*. Sage publications, 2017.
[39] I. Han, "Immersive virtual field trips in education: A mixed-methods study on elementary students' presence and perceived learning," *Br. J. Educ. Technol.*, vol. 51, no. 2, pp. 420–435, 2020.

KEMENTERIAN PENDIDIKAN, KEBUDAYAAN, RISET DAN TEKNOLOGI
UNIVERSITAS PENDIDIKAN INDONESIA
FAKULTAS PENDIDIKAN MATEMATIKA DAN ILMU PENGETAHUAN ALAM
Jl. Dr. Setiabudhi Nomor 229 Bandung 40154
Telepon/Faks. (022) 2001108, 2013163 Ext 4632
Laman: fpmipa.upi.edu; surel/e-mail: fpmipa@upi.edu

Nomor	:	B-11849/UN40.A4.1/PT.01.04/2024
Lampiran	:	1 berkas
Perihal	:	*Permohonan Izin Penelitian*

Yth.
SMAS Plus Permata Insani Islamic School, PERUM VILLA PERMATA BLOG G1, SINDANG SARI, KEC. PASAR KEMIS, KAB. TANGERANG, PROV. BANTEN

Bersama ini kami sampaikan bahwa mahasiswa Fakultas Pendidikan Matematika dan Ilmu Pengetahuan Alam Universitas Pendidikan Indonesia,

Nama	:	MOHD ZAIDI BIN AMIRUDDIN
NIM	:	2310958
Program Studi	:	PENDIDIKAN ILMU PENGETAHUAN ALAM/S2
Dosen Pembimbing	:	Dr. Achmad Samsudin, M.Pd. Prof. Dr. Andi Suhandi, M.Si.

bermaksud untuk melaksanakan penelitian di tempat yang Bapak/Ibu pimpin dalam rangka penyusunan Tesis dengan judul "**PENGEMBANGAN REFUTATIONAL TEXTS BERBANTUAN AUGMENTED REALITY (RefTaR) UNTUK MENGUBAH KONSEPSI DAN MODEL MENTAL PADA MATERI LISTRIK STATIS**". Sebagai bahan pertimbangan Bapak/Ibu, bersama ini kami sampaikan,

1. Proposal penelitian/deskripsi penelitian 1 eksemplar;
2. Fotokopi KTM 1 lembar

Besar harapan kami, Bapak/Ibu dapat memberikan izin kepada mahasiswa bersangkutan untuk melakukan kegiatan tersebut.
Atas perhatian dan bantuan Bapak/Ibu, kami ucapkan terima kasih

Bandung, 22 Oktober 2024

Wakil Dekan Bidang Akademik,

KEMENTERIAN PENDIDIKAN, KEBUDAYAAN, RISET DAN TEKNOLOGI UNIVERSITAS PENDIDIKAN INDONESIA FPMIPA

Dr. rer.nat. Adi Rahmat, M.Si.
NIP. 196512301992021001

KEMENTERIAN PENDIDIKAN, KEBUDAYAAN, RISET DAN TEKNOLOGI
UNIVERSITAS PENDIDIKAN INDONESIA
FAKULTAS PENDIDIKAN MATEMATIKA DAN ILMU PENGETAHUAN ALAM
Jl. Dr. Setiabudhi Nomor 229 Bandung 40154
Telepon/Faks. (022) 2001108, 2013163 Ext 4632
Laman: fpmipa.upi.edu; surel/e-mail: fpmipa@upi.edu

Nomor : 480/UN40.F4.D1/KM/2023
Lampiran : 1 berkas
Perihal : *Permohonan Izin Penelitian*

Yth.
SMAS PLUS PERMATA INSANI ISLAMIC SCHOOL, PERUM VILLA PERMATA BLOG G1, SINDANG SARI, KEC. PASAR KEMIS, KAB. TANGERANG, PROV. BANTEN

Bersama ini kami sampaikan bahwa mahasiswa Fakultas Pendidikan Matematika dan Ilmu Pengetahuan Alam Universitas Pendidikan Indonesia,

Nama	:	MOHD ZAIDI BIN AMIRUDDIN
NIM	:	2310958
Program Studi	:	PENDIDIKAN ILMU PENGETAHUAN ALAM/S2
Dosen Pembimbing	:	Dr. Achmad Samsudin, M.Pd. Prof. Dr. Andi Suhandi, M.Si.

bermaksud untuk melaksanakan penelitian di tempat yang Bapak/Ibu pimpin dalam rangka penyusunan Tesis dengan judul "**PENGEMBANGAN REFUTATIONAL TEXTS WITH AUGMENTED REALITY (RefTaR) BERBANTUAN ARDUINO-UNO UNTUK MENGUBAH KONSEPSI DAN MODEL MENTAL PADA KONSEP GERAK LURUS**". Sebagai bahan pertimbangan Bapak/Ibu, bersama ini kami sampaikan,

1. Proposal penelitian/deskripsi penelitian 1 eksemplar;
2. Fotokopi KTM 1 lembar

Besar harapan kami, Bapak/Ibu dapat memberikan izin kepada mahasiswa bersangkutan untuk melakukan kegiatan tersebut.
Atas perhatian dan bantuan Bapak/Ibu, kami ucapkan terima kasih

Bandung, 17 Januari 2024

Wakil Dekan Bidang Akademik,

KEMENTERIAN PENDIDIKAN, KEBUDAYAAN, RISET DAN TEKNOLOGI
UNIVERSITAS PENDIDIKAN INDONESIA
FPMIPA

Dr. rer.nat. Adi Rahmat, M.Si.
NIP. 196512301992021001

Sekolah Menengah Atas Plus (SMA)
Permata Insani
Terakreditasi A (Unggul) Nomor : 104/BAN PDM/SK/2024
Permata Insani Islamic School

PERMATA INSANI
ISLAMIC SCHOOL

Sekretariat :
Komplek Perumahan VILLA PERMATA
Ps. Kemis - Tangerang, BANTEN
Telp. 089632073052 / 02159350234
Website : www.permatainsaniislamicschool.sch.id
E-mail : smaplus.permatainsani@gmail.com

SURAT KETERANGAN

233/YPP-PI/SMA PLUS/XII/2024

Dengan hormat,

Yang bertanda tangan dibawah ini

Nama	: **Hasan Firdaus, S.Kom., M.Pd**
Jabatan	: Kepala Sekolah

Menerangkan bahwa,

Nama lengkap	: Mohd Zaidi Bin Amiruddin
Tempat tanggal lahir	: Tawau, 5 Juni 2000
Perguruan tinggi	: Universitas Pendidikan Indonesia
Program Study	: Pascasarjana Pendidikan IPA
Nomor Induk Mahasiswa	: 2310958
Alamat	: Gang Gegersari No 30, RT 3/RW 1. Gegerkalong, Sukasari, Kota Bandung, Jawa Barat (40153)
No. Telephone/Whatsapp	: +6289653881924
Email	: mohdzaidi@upi.edu

Telah selesai melaksanakan penelitian di SMAS Plus Permata Insani Islamic School sebagai syarat menyelesaikan thesis yang berlangsung sejak tanggal 03 - 29 November 2024 dengan judul **"PENGEMBANGAN REFUTATIONAL TEXTS BERBANTUAN AUGMENTED REALITY (RefTaR) UNTUK MENGUBAH KONSEPSI DAN MODEL MENTAL PADA MATERI LISTRIK STATIS"**

Demikian surat ini dibuat dengan sebenar-benarnya dan untuk dapat digunakan sebagaimana mestinya.

Mengetahui,
Kepala Sekolah
SMA Plus Permata Insani Islamic School

DINAS PENDIDIKAN DAN KEBUDAYAAN PROVINSI BANTEN
SMA PLUS PERMATA INSANI ISLAMIC SCHOOL

Hasan Firdaus, S.Kom. M.Pd

Tembusan :

- *Arsip*

LEMBAR SOAL

Mata Pelajaran **:** Fisika (Gerak Lurus Beraturan)
Satuan Pendidikan : SMA (Sekolah Menengah Atas)

Nama :

Kelas :

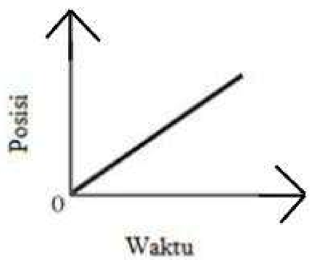
No presensi :

PETUNJUK:

a. Tulis nama, kelas, dan no presensi Anda pada lembar soal yang diberikan.
b. Kerjakanlah soal-soal berikut dengan sungguh-sungguh sesuai dengan pengetahuan dan pengalaman Anda.
c. Tes ini dilakukan dalam rangka untuk mengevaluasi dan mengetahui profil peserta didik terkait dengan materi gerak lurus beraturan.
d. Hindari jawaban dengan cara menebak atau dengan perkiraan dan jangan meminta jawaban dari teman, jawaban Anda mengambarkan apa yang Anda pikirkan secara personal.
e. Lembar soal dikumpulkan kembali setelah Anda selesai mengerjakan dalam selang waktu yang telah ditentukan.

Posisi, jarak dan perpindahan

1. Perhatikan grafik berikut!



Manakah pernafsiran terbaik dari garafik tersebut?

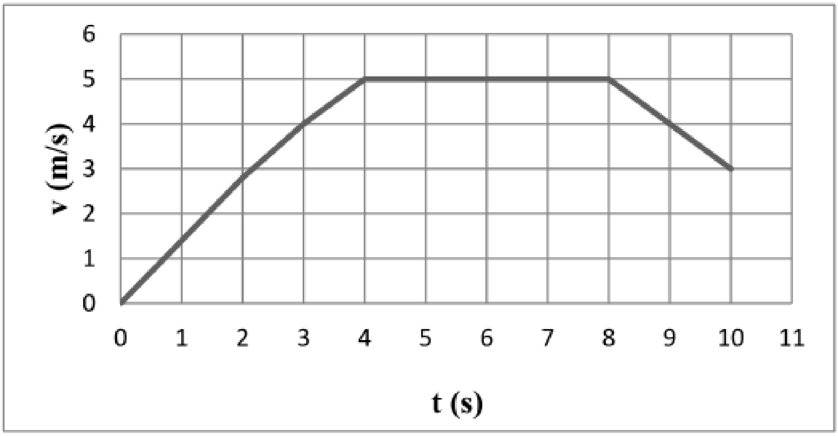
A. Objek bergerak dengan percepatan konstan tidak nol
B. Objek tidak bergerak
C. Objek bergerak dengan kecepatan yang meningkat seacara beraturan
D. Objek bergerak dengan kecepatan konstan
E. Objek bergerak dengan perceptan yang meningkat seacara beraturan

Tingkat keyakinan terhadap pilihan jawaban

1. Yakin 2. Tidak Yakin

Alasan terhadap pilihan jawaban

2. Sebuah elevator bergerak dari lantai satu ke lantai dua belas sebuah bangunan. Massa elevator tersebut adalah 1000 kg dan bergerak sepertyi yang ditunjukkan pada grafik kecepatan terhadap waktu di bawah ini.



Seberapa jauh elevator tersebut bergerak dalam kecepatan tiga detik pertama gerak?

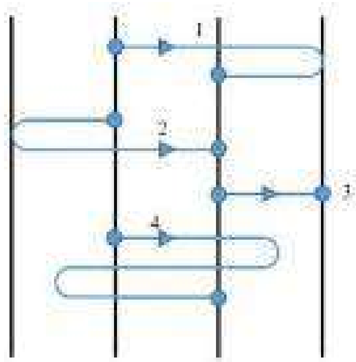
A. 0.75
B. 4.0
C. 6.0
D. 12.0
E. 1.33

Tingkat keyakinan terhadap pilihan jawaban

1. Yakin 2. Tidak Yakin

Alasan terhadap pilihan jawaban

3. Perhatikan Gambr berikut!



Pada keempat lintasan tersebut, terlihat bahwa objek bergerak dari titik awal ke titik akhir dalam interval waktu yang sama. Jarak antar lintasan garis lurus bernilai sama besar. Tentukan urutan objek berdasarkan urutan laju rata-rata yang akan dialami objek dari yang terbesar hingga kecepatan rata-rata terkecil..

A. 4,1,2,3
B. 4,3,2,1

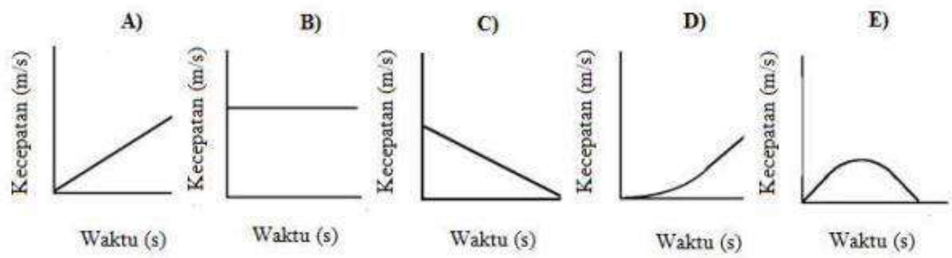
C. 1,2,3,4
D. 3,2,1,4
E. 4,1,3,2

Tingkat keyakinan terhadap pilihan jawaban
1. Yakin 2. Tidak Yakin

Alasan terhadap pilihan jawaban

Kecepatan, kelajuan, dan percepatan

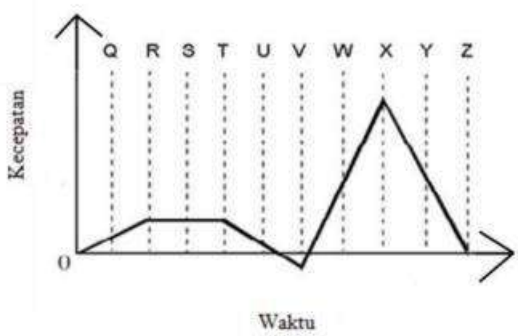
4. Berikut disajikan grafik kecepatan terhadap waktu. Semua sumbu mewakili skala yang sama. Objek mana yang mewakili perubahan posisi terbesar selama interval?



Tingkat keyakinan terhadap pilihan jawaban
1. Yakin 2. Tidak Yakin

Alasan terhadap pilihan jawaban

5. Perhatikan grafik berikut!

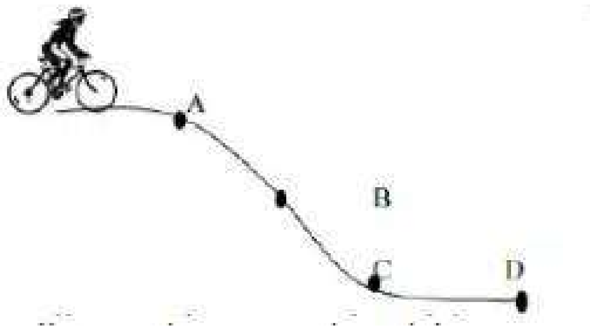


Berdasarkan dari garfik di atas, manakah yang memiliki perceptan yang paling kecil?
A. R ke T
B. T ke V
C. V
D. X
E. X ke Z

Tingkat keyakinan terhadap pilihan jawaban
1. Yakin   2. Tidak Yakin

Alasan terhadap pilihan jawaban

6. Seseorang sedang bersepada menuruni sebuah bukit yang bentuknya seperti pada gambar dibawah. Gaya gesek antara sepeda dengan lintasan bukit diabaikan.Apa yang terjadi dengan besarnya kecepatan dan percepatan pada saat orang tersebut menuruni bukitdari A=B?

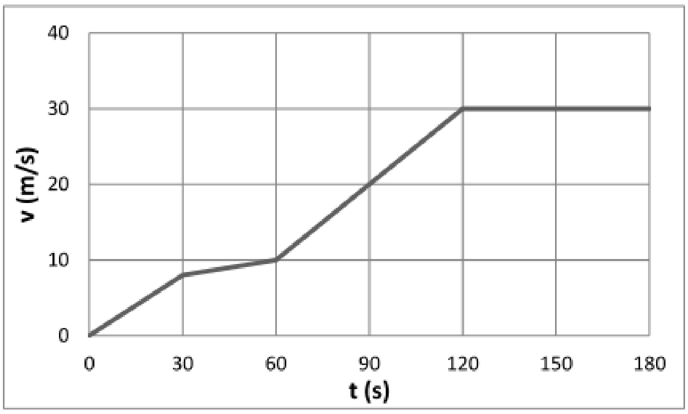


A. Kecepatan dan percepatan bertambah
B. Kecepatan dan percepatan berkurang
C. Kecepatan dan percepatan tetap
D. Kecepatan bertambah dan percepatan berkurang.
E. Kecepatan bertambah dan percepatan tetap

Tingkat keyakinan terhadap pilihan jawaban
1. Yakin   2. Tidak Yakin

Alasan terhadap pilihan jawaban

7. Grafik di bawah menunjukkan kecepatan sebagai fungsi waktu untuk mobil bermassa 1500 kg. Berapakah percepatan pada saat 90 s?



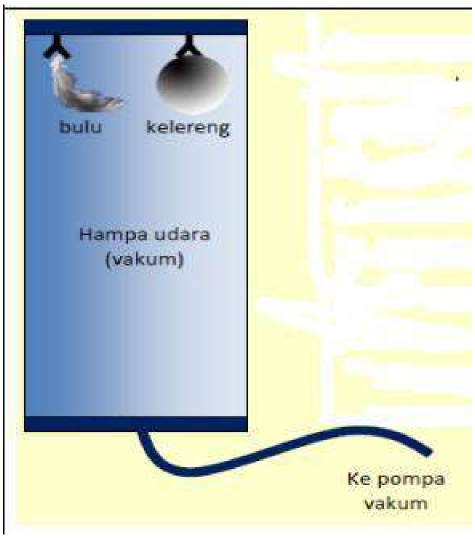
A. 0.22 m/s^2
B. 0.33 m/s^2
C. 1.0 m/s^2
D. 9.8 m/s^2
E. 20 m/s^2

Tingkat keyakinan terhadap pilihan jawaban
1. Yakin 2. Tidak Yakin

Alasan terhadap pilihan jawaban

Gerak jatuh bebas

8. Gambar di bawah memperlihatkan sehelai bulu dan sebutir kelereng yang dilepas bersamaan dari ketinggian yang sama di dalam tabung hampa udara.



. Pernyataan manakah yang benar?

A. Kedua benda jatuh sampai di dasar bersamaan
B. Bulu sampai di dasar lebih dulu
C. Kelereng sampai di dasar lebih dulu

D. Keduanya melayang-layang dan benda mana yang sampai di dasar lebih dulu tidak dapat ditentukan

Tingkat keyakinan terhadap pilihan jawaban
1. Yakin 2. Tidak Yakin

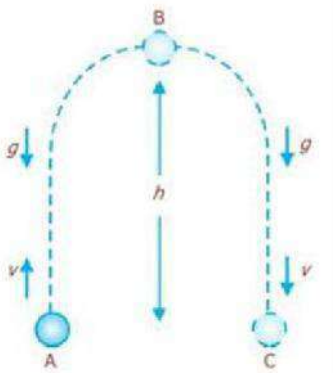
Alasan terhadap pilihan jawaban

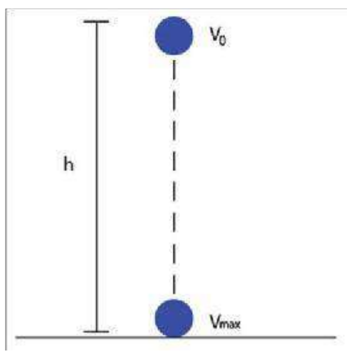
9. Apabila sebuah Kelapa jatuh bebas dari ketinggian tertentu maka kecepatan awal buah kelapa sebelum jatuh ke tanah ialah....
A. Bernilai negative
B. Bernilai Positif
C. Bernilai nol
D. Nilai kecepatan konstan
E. Nilai kecepatan sama dengan percepatan

Tingkat keyakinan terhadap pilihan jawaban
1. Yakin 2. Tidak Yakin

Alasan terhadap pilihan jawaban

10. Perbedaaan karakteristik dari gerak jatuh bebas dan gerak vertikal ke atas ditunjukkan pada gambar berikut.





Pernyataan manakah yang paling tepat….

A. Gerak vertikal ke atas kecepatan saat di puncak = 0 dan percepatannya a= -g. Sedangkan gerak jatuh bebas tidak memiliki kecepatan awal pada saat di titik puncak v_0=0; a= g
B. Gerak vertikal k etas kecepatan saat di puncak sama dan percepatannya a= -g. Sedangkan gerak jatuh bebas tidak memiliki kecepatan awal pada saat di titik puncak V_0= 0; a= -g

C. Gerak vertikal ke bawah kecepatannya saat di puncak = 0 dan percepatannya a= -g. Sedangkan gerak jatuh bebas tidak memiliki kecepatan awal pada saat di titik puncak V_0 = 0; a= -g.
D. Gerak vertikal ke atas kecepatan saat di puncak = 0 dan percepatannya a= g. Sedangkan gerak jagtuh bebas tidak memeiliki keceatan awal ada saat di titik puncak V_0 = 0; a = g.
E. Gerak vertikal ke atas kecepatan saat di puncak = 0 dan percepatannya a= g. Sedangkan gerak jatuh bebeas tidak memiliki keceoatan awal pada saat di titik puncak V_0 = 0; a = -g.

Tingkat keyakinan terhadap pilihan jawaban

1. Yakin 2. Tidak Yakin

Alasan terhadap pilihan jawaban

Jawaban

No	Jawaban	Alasan
1	D. Objek bergerak dengan kecepatan konstan	Objek mengalami keceptan konstan karena besar kevepatan dan aras kecepatan selalu konstan
2	D. 6.0	Perpindahan (peeubahan posisi) yang dialami oleh elevataor tersebut dapat dicari dengan luasan di bawah kurva
3	A. 4,1,2,3	Karena total lintasan yang ditempuh paling banyak adalah partikel 4 dan disusul dengan partikel 1 dan 2 yang memiliki panjang lintasan yang sama besar, dan yang terkecil panjang lintasan dari partikel 3. **Coba lengkapi nilai panjang lintasan yang ditempuh partikel 1, 2, 3, dan 4 tersebut!** $\Delta x_1 = a + a + a = 3a$ $\Delta x_2 = a + a + a = 3a$ $\Delta x_3 = a = a$ $\Delta x_4 = a + \frac{1}{2a} + \frac{1}{2a} + a + \frac{1}{2a} + \frac{1}{2a} + a = 5a$ **Kemudian, lengkapi rumus dari laju rata rata berikut!** $kecepatan = \frac{jarak}{waktu}$
4	B. Grafik	Perubahan posisi yang dialami oleh objek dapat ditentukan dengan luasan daerah di bawah kurva yang berbentuk bangun persegi panjang, maka dapat disimpulkan bahwa pada opsi B terlihat perubahan posisi yang paling besar.
5	E. X ke Z	Bnedan bergerak dengan perceatan paling kecil karena kemiringan sangat curam
6	A. Kecepatan dan percepatan bertamba	Karena semakin kebawah kecepatan semakin besar, waktunya semakin lama sehingga kecepatan dan percepatan semakin bertambah
7	B. 0,33 m/s^2	Percepatan yang dialami mobil merupakan gradien garius singgung grafik kecepatan v(t).
8	A. Kedua benda jatuh sampai di dasar bersamaan	Va = $v_0 \pm$ a. t Tidak dipengaruhi massa Ruang hampa = tidak ada gesekan
9	C. Bernilai Nol	Semua benda yang melakukan gerak jatuh bebas mendapatkan percepatan yang sama besar yaitu sebesar percepatan gravitasi bumi yang diberi simbol g. Disebut jatuh bebas karena gerak ini bebas dari adanya gaya dorong sehingga kecepatan awalnya nol
10	A. Gerak vertikal ke atas kecepatan saat di puncak = 0 dan percepatannya a= -g. Sedangkan gerak jatuh bebas tidak memiliki kecepatan awal pada saat di titik puncak v0=0; a= g	Karena pada saat di puncak benda akan berhenti sejenenak dan itu menandakan bawah kecepatan = 0 dan gerak perceotan berlawanan arah gravitasi bumi. Sedangkan keceoatn awal pada saat jatuh bebas sama dengan nol dan a = g karena searah dengan percepatan gravitasi

Social Sciences & Humanities Open

Revealing High School Students' Misconceptions Using 3-Tier Diagnostic Open-Ended Test (3-DOET) about Linear Motion: A Rasch Measurement Approach

--Manuscript Draft--

Manuscript Number:	SSHO-D-24-03178
Full Title:	Revealing High School Students' Misconceptions Using 3-Tier Diagnostic Open-Ended Test (3-DOET) about Linear Motion: A Rasch Measurement Approach
Article Type:	Review Article
Section/Category:	Education
Keywords:	Concept; Diagnostic test; gender; Misconceptions; Rasch analysis
Manuscript Region of Origin:	INDONESIA

Abstract:	This study develops and evaluates a 3-Tier Diagnostic Open-Ended Test (3-DOET) to diagnose misconceptions in linear motion among high school students. The 3-DOET is an advanced adaptation of Treagust's diagnostic test procedures and includes three tiers of assessment, a multiple-choice question (Tier-1), open-ended reasoning (Tier-2), and a confidence rating (Tier-3). The test was designed to capture students' conception of linear motion by examining their answers, reasoning, and confidence levels. This study involved 112 high school students 68 females (60.7%) and 44 males (39,3%) from East Java and Banten province, Indonesia. The data was analyzed using Microsoft Excel and WINSTEP software. Rasch analysis provided insights into item difficulty, reliability, and the test's unidimensionality. The results of this study state that students' conceptions are still dominated by the misconception category in each question item. In addition, Rasch's analysis indicated varying levels of understanding and misconceptions among students, with the test demonstrating good reliability and validity overall. The study highlights the effectiveness of the 3-DOET in identifying and addressing misconceptions and suggests areas for further research to enhance the diagnostic tool and its application across broader contexts.

Editor-in-Chief
Social Sciences & Humanities Open

Dear Editor-in-Chief,

I am pleased to submit our manuscript entitled "Revealing High School Students' Misconceptions Using 3-Tier Diagnostic Open-Ended Test (3-DOET) about Linear Motion: A Rasch Measurement Approach" for your consideration for publication in **Social Sciences & Humanities Open**

This study presents the development and evaluation of diagnostic tool, the 3-Tier Diagnostic Open-Ended Test (3-DOET), designed to assess high school students' misconceptions in linear motion. Our diagnostic tool builds on Treagust's well-established diagnostic test procedures by incorporating three tiers: multiple-choice questions, open-ended reasoning, and a confidence rating. This allows us to capture not only students' answers but also their reasoning processes and confidence levels. Using Rasch measurement, we evaluated the test's reliability, item difficulty, and unidimensionality, providing a robust statistical basis for interpreting students' understanding and misconceptions.
The results show that misconceptions dominate students' conceptions of linear motion, and the Rasch analysis offers valuable insights into varying levels of understanding. We believe that this tool can help educators better identify and address misconceptions in linear motion, making it a valuable resource for improving science education.
We hope that this manuscript will be of interest to the readers of **Social Sciences & Humanities Open** and contribute to ongoing research in science education, particularly in the development of diagnostic assessments and misconception research. We believe the findings will stimulate further studies in this area and support the refinement of educational tools to improve conceptual understanding.

Thank you for your consideration. We look forward to your feedback.

Kind Regards
Mohd Zaidi Bin Amiruddin
Student
Universitas Pendidikan Indonesia

Competing Interest

The authors declare no competing interests.

Revealing High School Students' Misconceptions Using 3-Tier Diagnostic Open-Ended Test (3-DOET) about Linear Motion: A Rasch Measurement Approach

Abstract

This study develops and evaluates a 3-Tier Diagnostic Open-Ended Test (3-DOET) to diagnose misconceptions in linear motion among high school students. The 3-DOET is an advanced adaptation of Treagust's diagnostic test procedures and includes three tiers of assessment, a multiple-choice question (Tier-1), open-ended reasoning (Tier-2), and a confidence rating (Tier-3). The test was designed to capture students' conception of linear motion by examining their answers, reasoning, and confidence levels. This study involved 112 high school students 68 females (60.7%) and 44 males (39,3%) from East Java and Banten province, Indonesia. The data was analyzed using Microsoft Excel and WINSTEP software. Rasch analysis provided insights into item difficulty, reliability, and the test's unidimensionality. The results of this study state that students' conceptions are still dominated by the misconception category in each question item. In addition, Rasch's analysis indicated varying levels of understanding and misconceptions among students, with the test demonstrating good reliability and validity overall. The study highlights the effectiveness of the 3-DOET in identifying and addressing misconceptions and suggests areas for further research to enhance the diagnostic tool and its application across broader contexts.

Keywords: Concept, Diagnostic test, Gender, Misconceptions, Rasch analysis

1. Introduction

Misconception is a wrong or inaccurate understanding of a concept that is not in line with the scientific concepts of experts. Misconceptions are something crucial to discuss because it has an impact on future learning, particularly learning that has concepts in it (e.g., physics, biology, and chemistry). There are two terms related to concepts, namely conception and misconception. Conception is a person's initial information or knowledge, while misconceptions are errors that occur when connecting the concepts that have been obtained (Kulgemeyer & Wittwer, 2023; Roth, 2013). Misconceptions can be an obstacle to a student's learning process because they build new understandings based on wrong concepts. Many theories have been developed related to misconceptions, such as conceptual change theory (Özdemir & Clark, 2007; Vosniadou, 2007), constructivism theory (Fosnot & Perry, 1996), and cognitive theory (Piaget, 1976). Researchers, lecturers, and teachers have shown an interest in exposing misconceptions in the focus of their fields of study. Mastery of scientific concepts is urgently needed and has been studied and is relevant in the current change in educational paradigm, especially in physics learning. This can be proven by a bibliometric analysis study with the keywords "Misconception" OR "Misconceptions" AND "Physics" presenting research trends on misconceptions in physics learning in Figure 1.

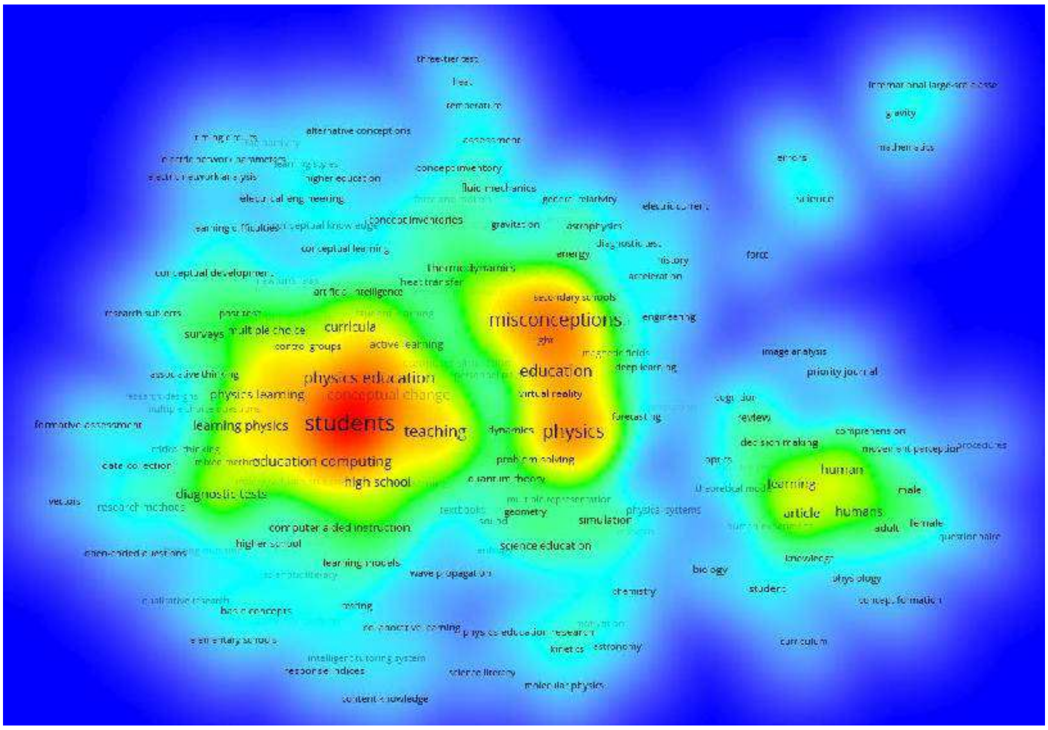


Figure 1. Trends in bibliometric analysis

Based on Figure 1, it is evident that the trend of research on misconceptions in physics learning continues to grow, with the most frequently appearing keywords indicating a close relationship between misconceptions, physics, students, and education. One form of instrument that can be used to reveal student misconceptions is through the tier-test instrument. According to Sesli and Kara (2012), a tier-test or multilevel test is an evaluation tool used to identify and understand student misconceptions in understanding certain concepts. Several previous studies have used tier tests to reveal misconceptions in physics material, ranging from two-tier (Ivanjek et al., 2021), three-tier (Taslidere, 2016), and four-tier (Kaltakci-Gurel et al., 2017). One of the reasons why it is important to reveal these misconceptions is that a good understanding of concepts can be a factor that predicts student results and achievement. In line to Binder et al. (2019) stated that a good understanding of concepts, principles, and abilities can be used in completing and determining academic achievements. Apart from that, mastering concepts can also facilitate the formation of mental models and learning processes that can be used in solving problems, especially in the context of physics learning (Didiş et al., 2014). Additionally, Indonesia is ranked lowest among 41 countries in the 2018 PISA report regarding student performance in science (OECD, 2020). This shows that many students in Indonesia face difficulties in understanding scientific concepts during the learning process.

Previous studies show several research efforts to identify misconceptions in certain contexts. Resbiantoro and Setiani (2022), examined 72 studies addressing misconceptions, Gurel et al. (2015) identified 273 articles about misconceptions, and Soeharto et al. (2019) also discovered 111 articles centered on student misconceptions in science. Several articles identify misconceptions using the three-tier test (e.g., Arslan et al., 2012; Assimi et al., 2022; Gurcay & Gulbas, 2015; Taslidere, 2016), but these articles only present the results of misconceptions in groups without representing individual student conceptions. In this study, in addition to presenting group misconceptions, it also presents individual misconceptions and students' ability to answer the tests given through Rasch analysis. Consequently, a three-tier open-ended test (3-DOET) was created, utilizing the Rasch measurement model to detect and assess the students' misconceptions. The Rasch measurement model was used because it can convert research instruments to interval scales and is able to measure the level of ability of persons and items.

1.1. Student Misconceptions

Student misconceptions have emerged as a significant issue in science education. There is considerable variation in young people's grasp of scientific concepts, with one prominent issue being 'student misconceptions' (Stefanidou & Skordoulis, 2017). Misconceptions among students can be classified into several types such as scientific knowledge, misconception (false positive or false negative), misconception, lucky gues or lack of confidence, and lack of knowledge (Arslan et al., 2012); sound understanding, partial understanding, partial understanding with specific misconception, specific misconceptions , and no understanding (Coştu, 2008); scientific conception, lack of knowledge, false positive, false negative, misconception, mistake (Gurel et al., 2015). Conceptual misconceptions are confusing and incorrect understandings that students develop when they construct knowledge based on scientific concepts during the learning process (Keeley, 2012; Peşman & Eryılmaz, 2010). Students may find it difficult to distinguish between distance and displacement as these two concepts often look similar in the measurement of movement, but are different by definition. Preconceptions are common ideas formed by students from their personal life experiences

(Cartrette & Melroe-Lehrman, 2012). It was observed that students faced difficulties in determining various quantities, such as velocity from position-time graphs, acceleration and displacement from velocity-time graphs, and changes in velocity from acceleration-time graphs, among others (Beichner, 1994). Misconceptions occur due to the everyday use of words that have different scientific meanings. For instance, students may confuse mass and weight, velocity and speed, assuming they are the same (Liu & Fang, 2016; Samsudin et al., 2021). This study focuses on revealing students' misconceptions about linear motion material.

In previous study, numerous studies have focused on student misconceptions in learning physics, as these misconceptions are often resistant to change, persistent, and deeply rooted in certain physics concepts (Eshetu & Alemu, 2018; Kaltakci-Gurel et al., 2017; Potvin & Cyr, 2017; Topalsan & Bayram, 2019). On the other hand, when students have misconceptions about learning physics, they struggle to grasp more advanced physics concepts. In addtion, misconceptions can cause students to have difficulty learning material related to concepts in science subjects, especially in physics subjects. This study aims to reveal students' misconceptions about linear motion, as it can assist both students and teachers in gaining a clearer understanding of the related learning concepts and material.

1.2. Three-tier diagnostic instrument test

In recent years from 2015 to 2019, multi-tier diagnostic tests have become a widely used assessment tool for identifying student misconceptions across various research areas (Soeharto et al., 2019). The three-tier test is a further development of the two-tier test. The first tier assesses students' conceptions in the form of question content, the second tier consists of reasons which can be in the form of multiple choice options or open-ended , and the third tier assesses the level of self-confidence can be in the form of sure or not sure and a Likert scale (Cetin-Dindar & Geban, 2011; Korkmaz et al., 2019; Sun, 2009). In this research, the first tier of the question is designed to address students' common misconceptions about linear motion and will assess their content knowledge. The second tier is an open-ended format, exploring students' reasoning related to scientific concepts and potential alternative ideas. At the third tier, students will provide responses that indicate whether they are confident or unsure about their answers. Open-ended questions were created at Tier-2 as a basis for uncovering students' misconceptions personally.

The literature review shows that the use of 3-tier has been done in several previous research topics in phyics such as, heat and temperature test (Eryilmaz, 2010), simple electric circuit (Peşman & Eryılmaz, 2010), wave (Caleon & Subramaniam, 2010), circular motion (Kızılcık & Güneş, 2011), and gravity (Kaltakci & Didis, 2007). The three-tier test proved to be able to reveal students' conceptions related to the material topics tested. However, the measurement of conceptions on linear motion material comprehensively has not been done using a three-tier test so that this research develops 3-DOET to measure student conceptions in detail.

1.3. Rasch measurement

The Rasch measurement model, developed by George Rasch, is based on interactions between items and individuals, as well as probability estimates. This model describes these interactions using mathematical equations. Individuals with higher abilities are expected to correctly answer items that are of lower difficulty (Andrich & Marais, 2019). In this measurement model, the probability is determined by both the difficulty of the item and the ability of the individual. Essentially, it reflects the differences between the difficulty of the item and the individual's skill level (Boone, 2016). In Rasch measurement, both a person's ability and item difficulty are measured using an interval scale known as the logit. The parameters for items and individuals are completely independent of each other (Planinic et al., 2019; Sumintono & Widhiarso, 2014a). This means that students' abilities in measurement remain consistent without being affected by the level of item difficulty, and the level of item difficulty does not change even though the abilities of students or test takers vary. In this study, the Rasch model was used to analyze a three-tier diagnostic test with coding categories such as sound understanding (3), partial understanding (2), misconception (1), and no conception (0).

On the other hand, the Rasch model can measure unidimensionality to show that the items in the instrument are used to measure the same aspect that can be seen from each other's raw variance explained by measures. On the other hand, it can also determine the reliability of the items based on the Cronbach alpha value. Rasch analysis was used in this study to overcome some of the limitations of Classical Test Theory (CTT). CTT have four limitations in explaining the measurement model: (i) it uses ordinal data instead of interval (logit) scales; (ii) items and individuals in the measurements been interdependent; (iii) the reliability and validity of measurement instruments are heavily sampled-dependent; (iv) it relies on group-centered statistics, which are not ideal for explaining individual respondent measurements. (Barbic & Cano, 2016). Rasch analysis can map ability levels, items difficulty, gender and domicile bias, and dimensionality.

1.4. Research Purpose

This study developed a 3-DOET instrument to measure students' misconceptions in physics lessons on linear motion material. The questions answered in this research are as follows:

1. Is the 3-DOET developed valid and reliable based on Rasch measurement?
2. What is the profile of student misconceptions and item-person interactions in 3-DOET?
3. Is there an instrument bias based on gender-based on differential item functioning (DIF)?

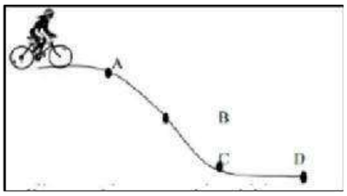
2. Method and Procedure

2.1. Development of the Test

The development of the 3-Tier diagnostic open-ended test (3-DOET) misconception test on linear motion is an advanced adapted version of the procedure described by Treagust (1988), namely to develop a diagnostic test. The development of 3-DOET was carried out through several main stages, namely: (1) Identifying misconceptions that often occur in linear motion, (2) Specifying sub-concepts in linear motion material, and (3) Developing 3-DOET. This 3-DOET is deliberately made open-ended because it is an initial measurement to reveal students' conceptions based on the answers and reasons they put forward. This is done to anticipate students' answer choices at Tier-1 because they are in the form of multiple choices or can be chosen directly. That way, through open-ended, students can express reasons for choosing answers in Tier-1 and followed by reasons and level of belief in Tier-2 and Tier-3.

The 3-DOET was developed by considering the results of a previous literature review related to the context of misconceptions about linear motion. The main objective of developing 3-DOET was carried out as a case study to obtain generalizations at different research sites for conducting further research. Questions on 3-DOET consist of 2 sub-concepts of linear motion with each concept being a sub-concept (1) position, distance, displacement, and sub-concept (2) speed, velocity, and acceleration. The number of questions presented on the concept of linear motion is 6 questions with details of 3 questions (sub-1) and 3 questions (sub-2). The list of misconceptions and alternatives is presented in Table 1. An example of the questions from 3-DOET is presented in Figure 2.

A person is cycling down a hill that is shaped as shown below. The friction force between the bicycle and the hill is ignored. What happens to the velocity and acceleration as the person descends the hill from A to B?



A. Velocity and acceleration increase *
B. Velocity and acceleration decrease
C. Velocity and acceleration remain fixed
D. Velocity increases and acceleration decreases
E. Velocity increases and acceleration remains constant

Reasons for answer choices

....

Confidence level of answer choices
1. Sure 2. Not sure

Figure 2. An example of 3-DOET

Table 1. List of misconceptions and alternative set measuring

Sub-concept	Misconception	Alternative question
Position, distance, displacement	Distance equals displacement, and the greater the distance, the greater the velocity	1.a, 1.b, 1.c. 2.a, 2.b, 2.c 3.a, 3.b, 3.c
	Objects in front of other objects move with greater speed regardless of the distance traveled	
	If there are many paths, the speed will decrease	
Speed, velocity, acceleration	A negative velocity indicates that the object is at rest	4.a, 4.b, 4.c. 5.a, 5.b, 5.c. 6.a, 6.b, 6.c.
	All objects whose acceleration is zero are only objects at rest	
	If the velocity of an object is equal to zero, then the acceleration is also zero	

Answers in Tier-1 are the content level are multiple choice. Then in Tier-2 items is open-ended and in Tier-3 is closed (choice sure or not sure). The form of the example question given is shown in Figure 2.

2.2. Participants

A total of 112 students from three secondary schools originating from two provinces, namely East Java and Banten, Indonesia. Participants consist of 68 females (60.7%) and 44 males (39.3%) with student ages ranging from 14-18 years with calcification such as grade 10 (n=11), grade 11 (n=8), and grade 12 (n=93). In addition, the tests were conducted at 3 different levels because the participants had already learnt about the physics of linear motion at the junior high school level and at the senior high school level in the first year entered school. The calls used for people from the province of East Java are male (Mas) and female (Mbak), while for Banten province there are no special calls, except for following the West Java regional calls for male (Akang) and female (Teteh). The location map between East Java and Banten is presented in Figure 3.

Figure 3. Map of the location research

2.3. Data Collection and Analysis

3-DOET was tested on high school students (10th, 11th, and 12th grades). The time given to answer these questions is around 25-30 minutes. The score obtained at Tier-1 if correct is 1 and if incorrect, is given the code M-1. For Tier-2, if it is correct is 1 and if it is incorrect it coded M-2. Then for Tier-3, the level of confidence is determined based on students' conceptions referring to the total tiers they have worked on. All categories presented in first, second, and three tiers are presented in Figure 4.

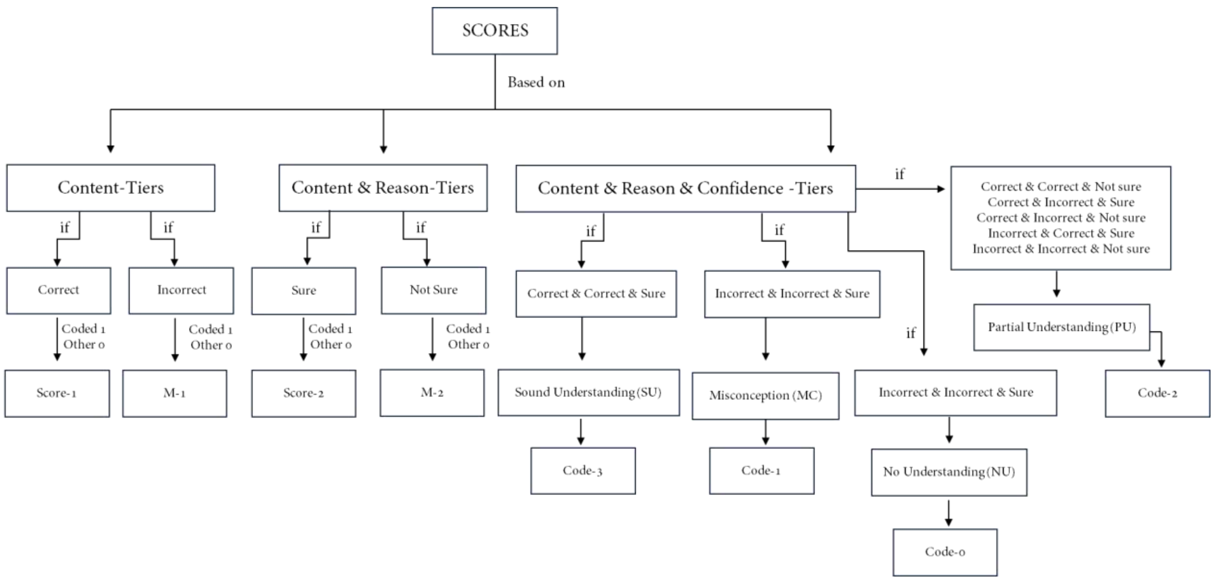


Figure 4. The diagram of the coding and scoring procedure

Tier-1, Tier-2, and Tier-3 are considered together to determine the total score that will be obtained. In 3-DOET for Tier-3, it is made closed (choice sure or not sure) that it can be the most meaningful reference in determining whether students are truly sure of the answers given through the reasons written in Tier-2. To determine in detail, the rubric used is an adaptation of the three-tier test criteria (Coştu, 2008; Gurel et al., 2015) which is presented in Table 2.

Table 2. Rubric of 3-DOET

No	Tier-1	Tier-2	Tier-3	Category	Score code
1	Correct	Correct	Sure	Sound Understanding (SU)	3
	Correct	Correct	Not sure		
	Correct	Incorrect	Sure		
2	Correct	Incorrect	Not sure	Partial Understanding (PU)	2
	Incorrect	Correct	Sure		
	Incorrect	Incorrect	Not sure		
3	Incorrect	Incorrect	Sure	Misconception (MC)	1
4	Incorrect	Incorrect	Not sure	No Understanding (NU)	0

The score code given in Table 2 is used for further analysis using Microsoft Excel and Rasch models with WINSTEP software. Data that has been processed using Microsoft Excel will be continued using WINSTEP to present the Rasch analysis results. The Rasch analysis output used is as follows:

2.3.1. Rasch Statistic

Rasch statistics include the Outfit mean square (MNSQ), Outfit Z-Standard value (ZSTD), Point Measure Correlation value (Pt Mean Corr), and Joint Maximum Likelihood Estimation (JMLE) Measure. These statistics are used to evaluate how well the data fits the model. The difficulty of each item is indicated by its JMLE Measure value (logit): a higher JMLE Measure signifies a more difficult item, while a lower value denotes an easier item. The appropriateness of question items (whether they are outliers or misfits) is determined based on these statistics. Data is considered a good fit when the Outfit mean square (MNSQ) value is within $0.5 < MNSQ < 1.5$, the Outfit Z-Standard value (ZSTD) falls between $-2.00 < ZSTD < 2.00$, and the Point Measure Correlation value (Pt Mean Corr) is within the range of $0.4 < Pt\ Measure < 0.85$ (Andrich & Marais, 2019; Chan et al., 2014).

2.3.2. Item and Person Reliability

The Rasch model measurement can also determine reliability regarding Cronbach's alpha, item reliability, and respondent reliability. Cronbach's alpha indicates the overall measurement of the interaction between responses and items. Additionally, Rasch analysis can assess the stability of respondents' answers through item and person reliability. The categories for Cronbach's alpha and item and person reliability are presented in Table 3 (Sumintono & Widhiarso, 2014b).

Table 3. Values category of cronbach’s alpha and item and person reliability

Cronbach’s Alpha			**Item and Person Reliability**	
Value	**Interpretation**		**Value**	**Interpretation**
$\alpha \geq 0.8$	Excellent		value $\geq$ 0.95	Excellent
$0.7 < \alpha \leq 0.8$	Very Good		0.91 < value $\leq$ 0.95	Very Good
$0.6 < \alpha \leq 0.7$	Good		0.81 < value $\leq$ 0.91	Good
$0.5 < \alpha \leq 0.6$	Moderate		0.68 < value $\leq$ 0.81	Moderate
$\alpha < 0.5$	Weak		value < 0.68	Weak

2.3.3. Udimensionality

The unidimensionality of the instrument is a parameter that can be used to measure the construction of the question instrument related to measuring what should be measured. Another aspect in measuring dimensionality is a variance that cannot be explained by the instrument (unexplained variance raw) which ideally does not exceed 15% (0.05). The criteria for unidimensionality are presented in Table 4 (Sumintono & Widhiarso, 2014a).

Table 4. Category of unidimensionality

Interpretation	Raw variance explained by measures
Special	>60%
In accordance	>40%
Fulfilled	>20%

2.3.4. Wright Map

One of the most valuable features of Rasch measurement is wright's maps, which enable researchers to quickly explain research results by illustrating individual abilities. (Wright, 1977). Wright maps depict the distribution of student ability and item difficulty in 3-DOET. The Wright map displays the distribution of individual and item scores based on logit values, using criteria such as the individual and item mean (M) and one (S) and two (T) standard deviations from the mean.

2.3.5. DIF (Differential Item Functioning)

Differential item functioning (DIF) is an analysis in the Rasch Model used to assess whether an item can consistently measure latent constructs across different groups. (Myers et al., 2006). DIF provides information about potential bias in respondent categories (e.g., gender and domicile). DIF analysis in this study evaluated 3-DOET items based on gender. The DIF testing criteria is if the probability value of the question is smaller than 5% (0.05), then the item can be said to be biased and can disadvantage certain groups.

3. Result

3.1. Is the 3-DOET developed valid and reliable based on Rasch measurement?

3.1.1. Summary Rasch measurement item characteristic

The characteristics of the 3-DOET items can be traced through a summary of the results of the Rasch measurement analysis which displays various statistics for the six items. Key metrics include total score, total count, Joint Maximum Likelihood Estimate (JMLE) measure, standard error (S.E.), infit mean square (MNSQ) and z-standardized (ZSTD) values, MNSQ and ZSTD outfit values, point-measure correlation (Corr.), expected correlation (Exp.), and percentage of correct matches. The details are presented in Table 5.

Table 5. Item characteristic 3-DOET

Item	Total score	JMLE measure	Model S.E.	It happens		Outfit		Pt Measure-AL		Exact Match	
				MNSQ	ZSTD	MNSQ	ZSTD	Corr.	Exp	Obs %	Exp %
Q4	97	1.18	0.17	1.04	0.32	1.05	0.34	0.48	0.40	62.5	66.2
Q2	110	0.83	0.16	**0.53**	**-3.48**	**0.53**	**-3.47**	0.47	0.42	79.5	67.4
Q5	140	0.12	0.15	1.08	0.60	1.09	0.61	0.36	0.47	56.3	61.9
Q3	154	-0.16	0.14	0.86	-0.99	0.80	-1.37	0.51	0.49	61.6	57.0
Q1	165	-0.36	0.13	1.09	0.72	1.16	1.10	0.63	0.50	52.7	55.7
Q6	242	-0.60	0.13	1.19	1.72	1.47	3.12	0.35	0.47	48.2	42.9
Mean	151.3	0.00	0.15	0.97	-0.18	1.02	0.06			60.1	58.5
P.SD	46.9	0.90	0.01	0.22	1.67	0.30	2.06			10.0	8.2

Table 5 depicts the characteristics of 3-DOET. The results of Table 5 present a total score ranging from 97 - 242, while the JMLE measure ranges from -0.60 - 1.18, indicating various difficulty levels of the questions. The value of infit and outfit MNSQ output for most of the items is within the acceptable range (0.5 to 1.5), indicating that the items generally fit the model, with Q2 having a value he enters and low MNSQ outfit and item 6 having a high MNSQ outfit value, so further investigation is needed. Apart from that, the average MNSQ infit value of 0.97 and the MNSQ outfit value of 1.02 indicate an overall good fit between the items and the Rasch model. Point-measure correlation values (Pt. Measure-Al), which ranged from 0.35 to 0.63, showed different levels of item discrimination, with some items correlating more strongly with the construct being measured. Observed (Obs%) and expected (Exp%) match percentages provide insight into the accuracy of item responses, with observed values generally close to expected values, indicating reliable measurements. However, differences in the percentage of correct matches for some items, such as item 2 with an observed percentage of 79.5% compared to an expected 67.4%, may indicate areas where the model could be improved. Standard deviations for various metrics indicate variability across items, which is common in the context of educational or psychological tests (Blanton & Jaccard, 2006; Putnick & Bornstein, 2016).

Additionally, through the Item Response Theory (IRT) framework, it can evaluate the performance of item matching by comparing theoretical expectations (red line) with the actual average score (blue cross line) presented in the graph Item Characteristic Curve (ICC) in Figure 5.

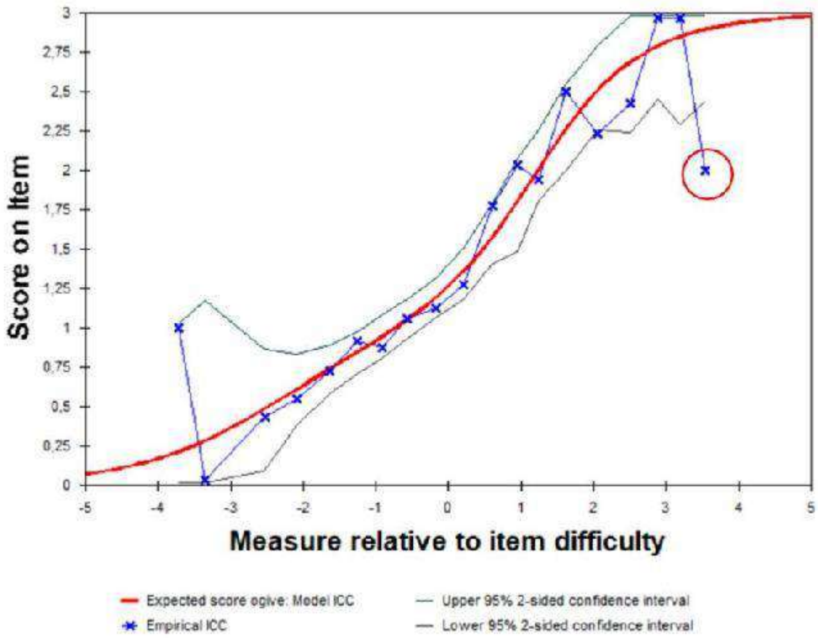


Figure 5. Item characteristic curve (ICC) measure

Based on Figure 5, respondents with moderate ability levels (between -2 and 2), the empirical ICC follows the ICC model, which shows good fit and accurate predictions. However, at lower ability levels (-5 to -3), the empirical scores deviate significantly, indicating that the model overestimates scores for respondents with less ability. Likewise, at higher ability levels (2 to 5), there is marked variability and some inconsistency in empirical values, with both overestimation and underperformance observed. The confidence intervals (green and black lines) depict insight into the uncertainty of the empirical scores. These intervals are wide at both lower and higher ability levels, indicating greater variability and less precision in these regions. In contrast, they narrow to moderate ability levels, showing increased confidence in the empirical scores. Overall, while the item fits the model well for moderate abilities, it shows a significant misfit at the extremes, suggesting a need for model adjustment or further investigation into the data quality at these ability levels.

3.1.2. Validity and reliability of 3-DOET

3-DOET was developed with psychometric properties measured by Rasch measurements. A total of 112 students and 6-item tests were processed using WINSTEP software to carry out Rasch analysis. The summary results of the item-person are presented in Figure 6.

SUMMARY OF 112 MEASURED Person

	TOTAL SCORE	COUNT	MEASURE	MODEL S.E.	INFIT MNSQ	INFIT ZSTD	OUTFIT MNSQ	OUTFIT ZSTD
MEAN	8.1	6.0	-.06	.63	1.00	-.16	1.02	-.13
SEM	.2	.0	.07	.00	.07	.13	.09	.12
P.SD	2.1	.0	.77	.05	.78	1.35	.95	1.29
S.SD	2.1	.0	.78	.05	.79	1.36	.95	1.29
MAX.	14.0	6.0	1.93	.82	3.57	2.78	6.81	3.82
MIN.	3.0	6.0	-2.53	.55	.07	-2.54	.09	-2.19

REAL RMSE .73 TRUE SD .27 SEPARATION .37 Person RELIABILITY .12
MODEL RMSE .63 TRUE SD .45 SEPARATION .72 Person RELIABILITY .34
S.E. OF Person MEAN = .07

Person RAW SCORE-TO-MEASURE CORRELATION = .99
CRONBACH ALPHA (KR-20) Person RAW SCORE "TEST" RELIABILITY = .28 SEM = 1.77
STANDARDIZED (50 ITEM) RELIABILITY = .81

SUMMARY OF 6 MEASURED Item

	TOTAL SCORE	COUNT	MEASURE	MODEL S.E.	INFIT MNSQ	INFIT ZSTD	OUTFIT MNSQ	OUTFIT ZSTD
MEAN	151.3	112.0	.00	.15	.97	-.18	1.02	.06
SEM	21.0	.0	.40	.01	.10	.75	.13	.92
P.SD	46.9	.0	.90	.01	.22	1.67	.30	2.06
S.SD	51.4	.0	.98	.02	.24	1.83	.32	2.25
MAX.	242.0	112.0	1.18	.17	1.19	1.72	1.47	3.12
MIN.	97.0	112.0	-1.60	.13	.53	-3.48	.53	-3.47

REAL RMSE .15 TRUE SD .88 SEPARATION 5.86 Item RELIABILITY .97
MODEL RMSE .15 TRUE SD .88 SEPARATION 6.03 Item RELIABILITY .97
S.E. OF Item MEAN = .40

Item RAW SCORE-TO-MEASURE CORRELATION = -.99

Figure 6. Summary of item-person

Figure 6 depicts a summary of person and item measurements for a dataset with 112 measured persons and 6 measured items. The analysis of the person's measurements reveals significant issues with reliability. The person reliability scores are notably low, with a real reliability of 0.12 (weak). This indicates low consistency in direct measurement, implying that the test may not effectively differentiate individual abilities. The cronbach's alpha (KR-20) for person raw score test reliability is only 0.28, which is well below acceptable thresholds, indicating poor internal consistency of the test items relating to the persons. Additionally, the high standard error of measurement (SEM) of 1.77 further underscores the unreliability, showing that the observed score is significantly affected by measurement error.

On the other hand, one of the things that can be used to determine and evaluate whether the instrument that has been developed has measured what should be measured can be known from the unidimensionality results presented (Segars, 1997). Table 6 provides the results of the unidimensionality of the 3-DOET which presents a raw variance explained measure of 41.6%. This shows that the minimum dimensionality requirement of 20% can be met. In this way, it can be concluded that 3-DOET is able to measure what it should measure but the answers from people are inconsistent, which has an impact on the cronbach alpha value.

Table 6. Dimensionality of 3-DOET

	Eigenvalue	**Observed** percentage of the total variance	**Observed** percentage of unexplained variance	**Expected** percentage of the total variance
Total raw variance in observations =	191.7199	100.0%		100.0%
Raw variance explained by measures =	79.7199	**41.6%***		41.1%
Raw variance explained by persons =	33.4019	17.4%		17.2%
Raw Variance explained by items =	46.3180	24.2%		23.9%
Raw unexplained variance (total) =	112.0000	58.4%	100.0%	58.9%
Unexplained variance in 1st contrast	41.3026	21.5%	36.9%	
Unexplained variance in 2nd contrast	26.6857	13.9%	23.8%	
Unexplained variance in 3rd contrast	21.1698	11.0%	18.9%	
Unexplained variance in 4th contrast	14.1383	7.4%	12.6%	
Unexplained variance in 5th contrast	8.7041	4.5%	7.8%	

3.2. What is the profile of students' misconceptions when learning linear motion physics?

The 112 students, their conceptions are certainly different from each other. Mapping students' conceptions is crucial to trace their understanding regarding the concepts tested in each question item. A total of 6 questions were asked and the results of the students' conceptions were presented in Table 7.

Table 7. Students' conception profile

Qs	Category	Students Code	f	Percentage
1	Sound Understanding	003ME,006FT,008FE,022FW,023FW,028FW,031FW,047MW,076MW,078FW,080FW,082MW,085MW,086FW,087FW,093FW,098FW,099FW,100FW,103MW,104FW,111MW	22	19,64%
	Partial Understanding	001ME,005FE,007MT,009ME,010ME,014MW,020MT,029FW,030MW,032FT,048MW,077MW,079MW,081FW,083FW,084FW,088MW,089MW,105FW,106FW,110FW,112MT	22	19,64%
	Misconception	002FE,004ME,011FT,012MW,013MW,015FW,016MW,018FW,019MW,021FT,024FT,026FT,033FW,034FW,035MW,036FW,039FW,040FW,042MW,043MW,044MW,046MW,049FW,050FW,051FW,052FW,053MW,054MW,055FW,056FW,057FW,058FW,060FW,061FW,063FW,065FW,066FW,067FW,068FW,069FW,070FW,071FW,072FW,073FW,074FW,075MW,090MW,091MW,092MW,095MW,096FW,097FW,101FW,102MW,107MW	**55**	**49,11%**
	No Understanding	017FW,025FT,027MT,037FW,038FW,041FW,045MW,059FW,062FW,064FW,094MW,108MW,109MW	13	11,61%
2	Sound Understanding	003ME	1	0,89%

Qs	Category	Students Code	f	Percentage
	Partial Understanding	004ME,005FE,006FT,007MT,008FE,009ME,016MW,017FW,043MW	9	8,04%
	Misconception	002FE,010ME,012MW,013MW,014MW,015FW,018FW,019MW,020MT,021FT,022FW,023FW,024FT,025FT,026FT,028FW,029FW,030MW,031FW,032FT,033FW,034FW,035MW,037FW,042MW,047MW,048MW,049FW,050FW,051FW,052FW,053MW,054MW,055FW,056FW,057FW,058FW,059FW,060FW,061FW,062FW,063FW,064FW,065FW,066FW,067FW,068FW,069FW,070FW,071FW,072FW,073FW,074FW,075MW,076MW,077MW,078FW,079MW,080FW,081FW,083FW,084FW,085MW,086FW,087FW,088MW,089MW,090MW,091MW,092MW,093FW,094MW,095MW,096FW,097FW,098FW,099FW,100FW,101FW,102MW,104FW,105FW,106FW,107MW,108MW,109MW,111MW,112MT	**89**	**79,46%**
	No Understanding	001ME,011FT,027MT,036FW,038FW,039FW,040FW,041FW,044MW,045MW,046MW,082MW,103MW,110FW	13	11,61%
3	Sound Understanding	003ME,004ME,020MT,022FW,023FW,028FW,030MW,031FW,050FW,051FW,052FW,077MW,079MW,085MW,104FW	15	13,39%

Qs	Category	Students Code	f	Percentage
	Partial Understanding	001ME,002FE,005FE,007MT,008FE,009ME,010ME,027MT,029FW,032FT,038FW,040FW,041FW,049FW,112MT	15	13,39%
	Misconception	006FT,012MW,013MW,014MW,015FW,016MW,017FW,018FW,019MW,021FT,022FW,024FT,025FT,026FT,033FW,034FW,035MW,036FW,039FW,042MW,043MW,044MW,046MW,047MW,048MW,053MW,054MW,055FW,056FW,057FW,058FW,059FW,060FW,061FW,062FW,063FW,064FW,065FW,066FW,067FW,068FW,069FW,070FW,071FW,072FW,073FW,074FW,075MW,076MW,078FW,080FW,081FW,082MW,083FW,084FW,086FW,087FW,088MW,089MW,090MW,091MW,092MW,093FW,094MW,095MW,096FW,097FW,098FW,099FW,100FW,101FW,102MW,103MW,105FW,106FW,107MW,108MW,109MW,110FW,111MW	**79**	**70,54%**
	No Understanding	011FT,037FW,045MW	3	2,68%
4	Sound Understanding	003ME,054MW,099FW	3	2,68%
	Partial Understanding	002FE,004ME,006FT,007MT,009ME,010ME,021FT,028FW,069FW	9	8,04%

Qs	Category	Students Code	f	Percentage
	Misconception	001ME,005FE,008FE,011FT,012MW,013MW,014MW,015FW,016MW,019MW,020MT,022FW,023FW,024FT,026FT,030MW,031FW,032FT,034FW,035MW,036FW,037FW,039FW,042MW,044MW,045MW,046MW,047MW,053MW,056FW,057FW,058FW,059FW,061FW,063FW,065FW,067FW,068FW,073FW,074FW,075MW,076MW,077MW,078FW,079MW,081FW,082MW,083FW,084FW,085MW,086FW,087FW,088MW,091MW,092MW,094MW,095MW,097FW,098FW,100FW,101FW,102MW,104FW,105FW,107MW,108MW,109MW,110FW,112MT	**70**	**62,50%**
	No Understanding	017FW,018FW,025FT,027MT,029FW,033FW,038FW,040FW,041FW,043MW,048MW,049FW,050FW,051FW,052FW,060FW,062FW,064FW,066FW,070FW,071FW,072FW,080FW,089MW,090MW,093FW,096FW,103MW,106FW,111MW	30	26,79%
5	Sound Understanding	016MW,021FT,056FW,093FW,096FW,098FW,099FW,100FW	8	7,14%
	Partial Understanding	001ME,004ME,009ME,011FT,036FW,039FW,045MW,059FW,060FW,061FW,062FW,064FW,065FW,067FW,068FW,090MW,105FW,106FW,108MW,110FW,111MW	22	19,64%
	Misconception	003ME,005FE,006FT,007MT,008FE,012MW,013MW,014MW,015FW,018FW,019MW,020MT,022FW,023FW,024FT,025FT,026FT,027MT,028FW,030MW,031FW,032FT,033FW,034FW,035MW,037FW,038FW,041FW,042MW,043MW,044MW,047MW,048MW,049FW,050FW,051FW,052FW,053MW,055FW,057FW,058F	**72**	**64,29%**

Qs	Category	Students Code	f	Percentage
		W,063FW,066FW,071FW,072FW,073FW,074FW,075MW,076MW,077MW,078FW,079MW,080FW,081FW,082MW,083FW,084FW,085MW,086FW,087FW,088MW,091MW,092MW,094MW,095MW,097FW,101FW,102MW,103MW,104FW,107MW,109MW,112MT		
	No Understanding	002FE,010ME,017FW,029FW,040FW,046MW,054MW,069FW,070FW,089MW	10	8,93%
6	Sound Understanding	006FT,007MT,013MW,017FW,021FT,022FW,023FW,024FT,025FT,027MT,028FW,029FW,030MW,031FW,032FT,033FW,034FW,038FW,040FW,041FW,050FW,051FW,052FW,055FW,057FW,059FW,061FW,062FW,064FW,071FW,073FW,074FW,075MW,086FW,093FW,097FW,098FW,099FW,100FW,101FW,104FW,105FW,106FW,109MW	**46**	**41,07%**
	Partial Understanding	005FE,011FT,012MW,014MW,015FW,016MW,018FW,019MW,020MT,035MW,036FW,037FW,039FW,042MW,044MW,045MW,047MW,048MW,049FW,053MW,056FW,058FW,060FW,063FW,065FW,066FW,067FW,068FW,069FW,070FW,072FW,076MW,078FW,080FW,082MW,088MW,089MW,090MW,091MW,092MW,094MW,095MW,107MW,108MW,110FW,111MW,112MT	44	39,29%
	Misconception	003ME,004ME,008FE,009ME,010ME,026FT,054MW,077MW,079MW,081FW,083FW,084FW,085MW,087FW,102MW	16	14,29%

Qs	Category	Students Code	f	Percentage
	No Understanding	001ME,002FE,043MW,046MW,096FW,103MW	6	5,36%

f: quantity, MT: Male 10th grade, FT: Female 10th grade, ME: Male 11th grade, FE: Female 11th grade, MW: Male 12th grade, FW: Female 12th grade

Table 7 presents students' conceptions of the 6 questions tested. Analysis of this data revealed students' understanding and misconceptions about linear motion material. The highest sound understanding (SU) category was Q6 with 46 students (41.07%) and the lowest was Q4 with 3 students (2.68%). Partial understanding (PU) was highest in Q1 with 22 students (19.64%) and lowest in Q4 with 9 students (8.04%). The highest misconception (MC) category was in Q2 with 89 students (79.46%) and the lowest was in Q6 with 16 students (14.29%). Then in the No understanding (NU) category, the highest was Q4 with 30 students (26.79%) and the lowest was Q6 with 6 students (5.36%). Based on Table 7, it can be seen that each question item is still dominated by the misconception category from Q1-Q6. This suggests that educational levels, whether grades 10, 11, or 12, still had misconceptions even though they have received linear motion material. Individual student abilities and the level of difficulty of the item can be traced through Rasch wright map analysis (Boone, 2016). The details are presented in Figure 7.

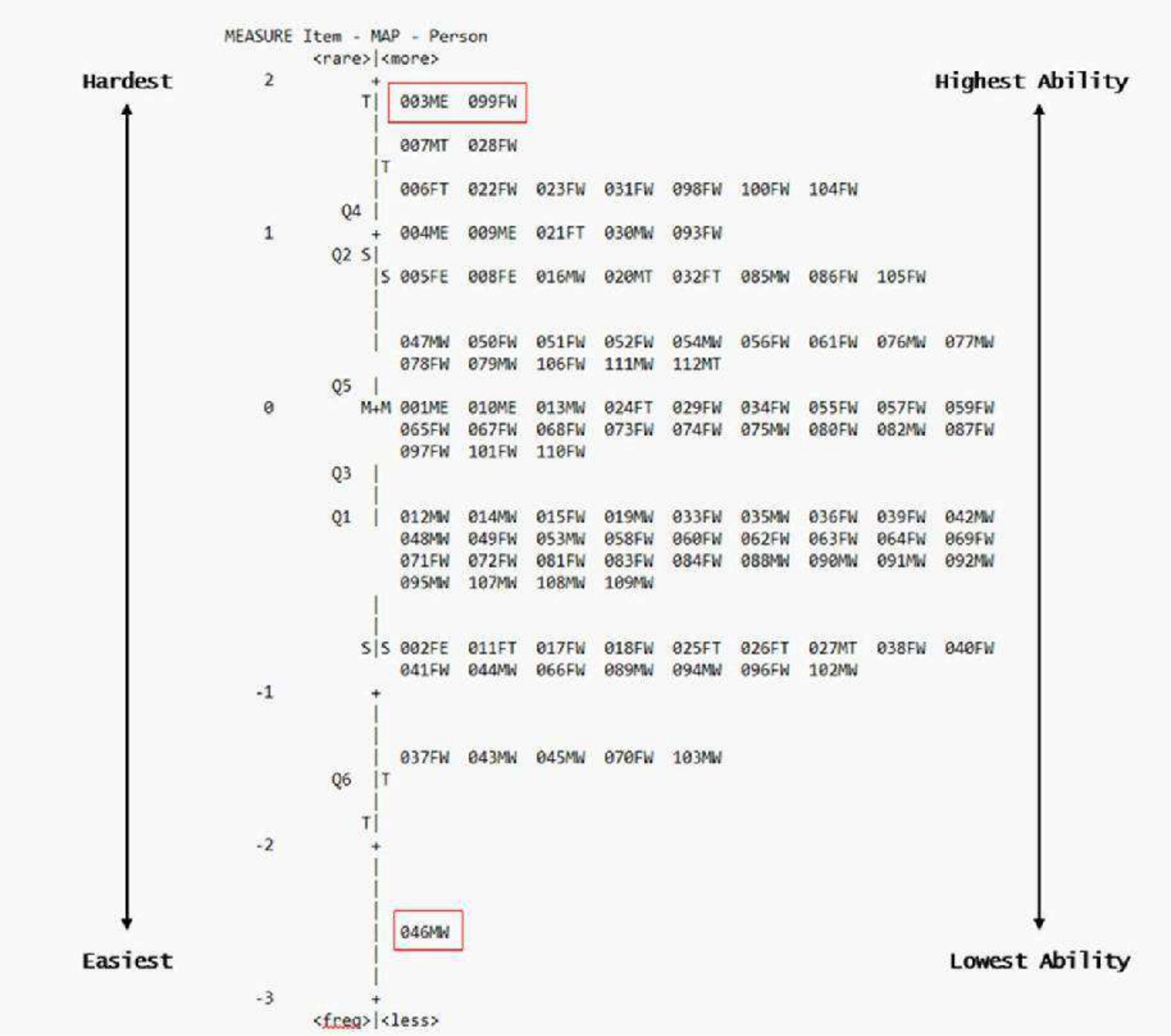


Figure 7. Wright map

Figure 7 depicts the distribution of students' abilities concerning the difficulty of the questions, highlighting how well different students perform on different questions. Figure 7 shows the level of difficulty of the questions, with the more difficult questions at the top and the easier questions at the bottom. Apart from that, it also shows the student's abilities, with higher abilities on the right and lower abilities on the left. Students 003ME and 099FW have the highest abilities and successfully answer the most difficult questions. On the other hand, 046MW students have the lowest abilities and are unable to answer the easiest question, namely Q6. Additionally, the wright map effectively identifies areas where students excel and struggle. Students like 007MT and 028FW, who are in the higher areas, can tackle more challenging material, showing that they can take advantage and answer the most difficult questions. In contrast, students who were at the bottom, such as 037FW and 103MW, need additional support and targeted interventions to improve their understanding of linear motion concepts.

3.3. Is there an instrument bias based on gender based on the differential item functioning (DIF) function?

When an item favors one individual or gender with a particular characteristic, it can be biased so that a person is disadvantaged. In this study, the DIF measurement was based on items regarding gender. As for the analysis that depicts DIF for gender is presented in Figure 8.

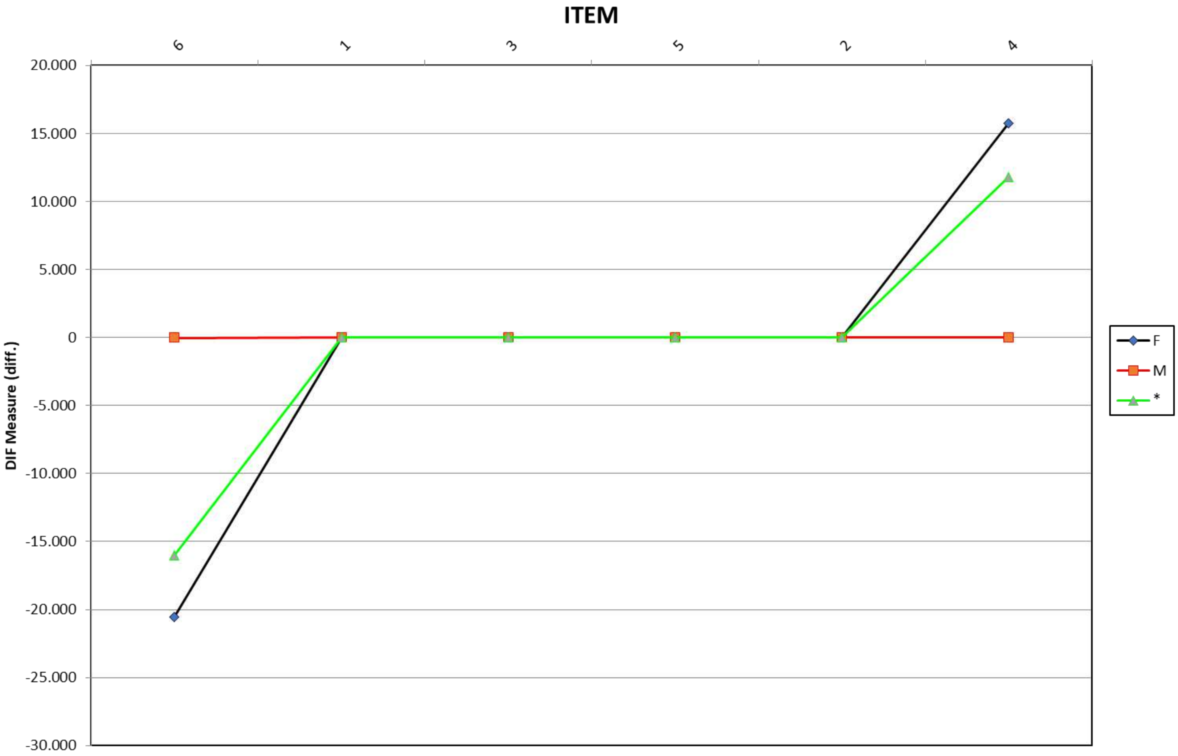


Figure 8. DIF based on gender

Figure 8 presents three lines with different colors. Blue represents males, red represents females, and green represents the ideal model from the results of Rasch's analysis. Items are said to be biased when they have a criterion probability <0.05. Based on Figure 8, two items are far from the ideal model, namely Q6 and Q4. This can be explored further by looking at the summary of probabilities presented in Table 8.

Table 8. Summary of probability DIF based on gender

Item Number	Name	Person Classes	Summary Dif Chi-Squared	D.F	PROB.	MNSQ	ZSTD
1	Q1	2	.7862	1	.3752	.7976	.32
2	Q2	2	1.1181	1	.2905	1.1324	.56
3	Q3	2	1.5895	1	.2074	1.6166	.84
4	Q4	2	7.6294	1	**.0057**	7.9824	2.59
5	Q5	2	.9132	1	.3393	.9184	.41
6	Q6	2	14.9434	1	**.0001**	16.8433	3.79

Table 8 depicts the six items Q1 to Q6 which reveal their significance and fit with the model. Items Q1, Q3, and Q5 show relatively low chi-square values, ranging from 0.7862 to 1.5895, with corresponding probabilities well above the >0.05 significance threshold. This indicates that the items do not show a bias towards gender, as indicated by a mean squared (MNSQ) value close to 1.0 and a standardized residual (ZSTD) close to zero. However, items Q4 and Q6 have high Chi-square values, respectively 7.6294 and 14.9434, and probabilities of 0.0057 and 0.0001 < 0.05. These values show a significant discrepancy, indicating that there is a bias towards gender. In addition, the discrepancy is supported by the high ZSTD values of 2.59 (Q4) and 3.79 (Q6), indicating that the items may measure different constructs or suffer from bias. Thus, Q4 and Q6 require further investigation and potential revision to ensure alignment with the overall model and accurately assess the constructs so that there is no detrimental bias.

4. Discussion

The results show that items Q1-Q6 have different characteristics from one another. The difficulty level of the questions can be seen from the logit value of the JMLS measure which shows that the most difficult question (Q4) has a logit value of 1.18 and the easiest question (Q6) has a logit value of -0.60. In line with state that the logit value describes the level of difficulty and ease of question items. Additionally, the MNSQ values were all within the acceptable range of 0.5<MNSQ< 1.5. Meanwhile, regarding the outfit value ZSTD, some items are outside the criteria, namely item Q2 (-3.47). Problem Q4 is the most difficult problem because it asks what if a car has a negative speed. Students assume that when this condition occurs, the velocity of a car does not exist. In fact, the speed is still there, it's just the opposite direction. Meanwhile, Q6 is the easiest question because the question only asks how the acceleration of an object when its speed is equal to zero.

Overall it is seen from the average value outfit (1.02) which has good agreement with the Rasch model. However, further investigation needs to be carried out to improve the quality of the items so that they fit with the Rasch model. Andrich and Marais (2019); Neumann et al. (2011), the item that matches a close model ideal can be a parameter in determining the similarity between observations with the expectations desired by the researchers. From the results characteristics 3-DOET indeed has the furthest gap between observation and expectation is item Q2. On the other hand, there is item Q3 but it is still within the acceptable category range. In addition, further analysis was carried out through the IRT framework (see Figure 5). ICC results compare expectation theoretical with the actual average scores with tolerance line limits. Although the overall measure relative to item difficulty is at the acceptable threshold, some items do not correspond to high proficiency. The items created must pay attention to suitability at both low and high levels so that the quality of the results research is trustworthy (Aladwani & Palvia, 2002; Elo et al., 2014). This is one of the importance of using Rasch analysis because it can map in as much detail as possible the characteristics of the question instrument being developed. That way, the quality of the measurement results can truly be trusted and is under what is desired.

Measuring validity and reliability through Rasch analysis can be seen from several points of view, namely, person reliability, item reliability, and cronbach alpha (KR-20) (Boone et al., 2013; Saidi & Siew, 2019). Person measurement results reliability, item reliability, and cronbach alpha have values of 0.12, 0.97, and 0.28 respectively. In contrast, the item's measurements demonstrate a much higher reliability. Both the real and modeled item reliability scores are 0.97 (excellent), indicating a strong consistency in how the items are measured across different persons. The separation index for items is also high (5.86 real, 6.03 model), suggesting that the items are well-distributed in terms of difficulty levels and are effectively discriminating between different levels of personal ability. The high item reliability indicates that the test items are functioning well, but the low person reliability points to potential issues such as sample heterogeneity, where the sample of persons may not be representative of the population, or an inadequate measurement model for the persons, indicating that the test may need to be revised or supplemented with additional items to improve its ability to differentiate between individuals effectively (Bond & Fox, 2013; Bornstein et al., 2013; Quaigrain & Arhin, 2017). On the other hand, hasil raw variance is explained by measures 41.6% (in accordance). 3-DOET has qualified criteria in the parameter of being able to measure what should be measured. According to Sumintono and Widhiarso (2014a), instruments which are developed and have a value of >40% are included in the category according to which means the instrument is valid in the appropriate category. However, the reliability of 3-DOET is still not good, because person's reliability is not consistent with the question items given so that cronbach's alpha value is in the weak category. One alternative that can be done is to increase the number of samples so that reliable instruments can be determined well (Kimberlin & Winterstein, 2008; Petersen et al., 2005).

3-DOET which has been developed with 6 questions is carried out trials on 112 students in Indonesia (East Java and Banten) with different levels, namely grades 10, 11, and 12. The students used as samples had received the material learning linear motion. The results served (see. Table 7) states that students' conceptions are dominated by category misconceptions (MC) both in Q1-Q6. Misconceptions can occur when participant students answer incorrectly in Tier-1 and Tier-2 but remain confident in the answer (Coştu, 2008; Gurel et al., 2015; Peşman & Eryılmaz, 2010). The misconceptions that occur do not only apply to students in grades 11 and 12 but also apply to grade 10 students who are still very fresh about linear motion material. This shows that the misconceptions that occur are not at the class level, but the understanding obtained by students is in accordance to the conceptions of the students (Chi, 2005; Samsudin et al., 2024). In addition, although students are correct in Tier-1, there are also mistakes in tier-2 (reason) until they are in partial category understanding (PU). Based on this, the open-ended (essay) in Tier-2 can be the key to seeing and understanding students related to the concepts they understand because it is the result of their own thinking. According to Rompayom et al. (2011), open-ended questions can determine students' deep understanding of what is being asked. Apart from that, you can understand and carry out mapping conception students so that they can become a reference for teachers to plan learning appropriately.

Futhermore, mapping conceptions is not enough, so information about students' personal abilities is needed. Matter that can be done through Rasch analysis is capable of measuring the level of ability and person-item difficulty (Boone & Scantlebury, 2006). The Wright map provides both (see. Figure 7) item and person measure. According to Boone et al. (2014), the wright map can provide detailed information regarding the level of difficulty of the questions and items level ability students to answer. That way, you can browse participant students who have high and low abilities. On the other hand, others can also see level the difficulty of the question is the most difficult to the easiest. Students with codes 003ME and 099FW are students with the highest abilities. However, the student was also not included in category sound understanding (SU) for all question items. Based on Rasch's analysis calculations, ability levels can be mapped to personal and difficult question items where students available to answer all the questions correctly. Besides that, students with code 046MW are in the category of misconception (MC) and no understanding (NU) on each question item. Thus, the mapping results on the wright map state that students' abilities are very low even under the easiest items. Individual mapping through Rasch analysis is very important because it is able to measure personally and not only groups (Akour, 2022; Wang, 2003). Therefore, learning more in an effort overcomes more misconceptions directed and mapped so make it easier teacher activities.

In addition, students' abilities and the level of difficulty of the questions can be further explored in relation to DIF to gender. Colton and Covert (2007); Kimberlin and Winterstein (2008), items of instruments developed need to be detected whether they contain a bias towards certain respondents so that they do not harm any of the parties involved. In this case, the DIF tested refers to the results of the 3-DOET development for gender. From these 6 items, it can be seen that Q6 and Q4 are on an abnormal path where both items have their own probability value < 0.05 which is the threshold for bias towards gender. This matter means there is a bias in items Q6 and Q4 towards gender. This can be explored further here student who has the ability to answer Q4 is dominated by female students, while Q6 has a low level of difficulty so students are able to answer. The factor why it is gender-biased is because these two items are the hardest and easiest items. Additionally, one of the become the factor is that this research is dominated by female students so which is influential against gender bias. According to Eagly et al. (1992); Moss-Racusin et al. (2012), when a research sample is unequal in terms of gender proportion, the results of the analysis may show biases that do not reflect the population as a whole and thus may be biased toward characteristics. Therefore, it is crucial to consider the gender distribution in the research sample to ensure that the findings obtained can be generalized more accurately and are not limited to one particular gender group.

5. Conclusion

Analysis of the validity and reliability of the 3-DOET using Rasch measurements shows that in general this test is by the Rasch model. However, several items show discrepancies, especially at extreme levels of ability. The validity of the test is classified as adequate with average he enters and MNSQ outfit within acceptable ranges, although some items, such as Q2, showed significant discrepancies. Meanwhile, test reliability is measured through reliability scores per person and Cronbach alpha, showing unsatisfactory results, with a low Cronbach's alpha and a high standard error of measurement, indicating significant uncertainty in the test results. In terms of dimensionality, the test showed fairly good unidimensionality with the explained variance exceeding the expected minimum of 41.6%. Apart from that, the student misconception profile shows that it is still dominated by the misconception (MC) category, thus indicating that there are challenges in understanding linear material motion.

Additionally, certain items show gender bias in items Q6 and Q4 where the value probability < 0.05. These results underscore the need for revisions to items that show bias and adjustments to the model to improve the accuracy and fairness of tests in measuring students' abilities more effectively.

5.1. Limitations and Suggestions

This study faces several limitations, which need to be considered in interpreting results and designing follow-up studies. First, the sample size limited to 112 students from three secondary schools in two provinces may not be fully representative of the wider student population in Indonesia. This can affect the generalization of findings regarding students' misconceptions and abilities in the context of linear motion. Second, the quality of the items in the 3-DOET does not fully conform to the Rasch model and displays gender bias that may affect the overall validity of the test. Third, the variability and reliability of 3-DOET may be affected by the uncertainty of the confidence level referring to Tier-1 and Tier-2 answers.

For further research, the 3-DOET can be developed and reused to assess misconceptions about linear motion not only for students, but also for inexperienced teachers, and experienced teachers in larger numbers and considering gender proportions. Additionally, educators can include 3-DOET in the list of tools that can be used to explore middle school students, aspiring science teachers, or aspiring physics teachers' conceptual understanding of linear motion. It is recommended that researchers develop a four-tier test to assess students' misconceptions in these areas of science in order to assess them in a more valid way. In addition, the level of confidence of the items only consists of two levels (sure and not sure) so it is necessary to revise the items by increasing the level of confidence to three, four, five, or six points in a Likert-type format.

Conflict of Interest

No potential conflict of interest was reported by the author

References

Akour, M. M. (2022). Rasch rating scale analysis of the survey of attitudes toward statistics. *EURASIA Journal of Mathematics, Science and Technology Education*, *18*(12), em2190.

Aladwani, A. M., & Palvia, P. C. (2002). Developing and validating an instrument for measuring user-perceived web quality. *Information & Management*, *39*(6), 467–476.

Andrich, D., & Marais, I. (2019). A course in Rasch measurement theory. *Measuring in the Educational, Social and Health Sciences*, *41*.

Arslan, H. O., Cigdemoglu, C., & Moseley, C. (2012). A three-tier diagnostic test to assess pre-service teachers' misconceptions about global warming, greenhouse effect, ozone layer depletion, and acid rain. *International Journal of Science Education*, *34*(11), 1667–1686.

Assimi, E., Janati Idrissi, R., Zerhane, R., & Boubih, S. (2022). The use of a three-tier diagnostic test to investigate conceptions related to cell biology concepts among pre-service teachers of life and earth sciences. *Journal of Biological Education*, 1–28.

Beichner, R. J. (1994). Testing student interpretation of kinematics graphs. *American Journal of Physics*, *62*(8), 750–762.

Binder, T., Sandmann, A., Sures, B., Friege, G., Theyssen, H., & Schmiemann, P. (2019). Assessing prior knowledge types as predictors of academic achievement in the introductory phase of biology and physics study programmes using logistic regression. *International Journal of STEM Education*, *6*, 1–14.

Blanton, H., & Jaccard, J. (2006). Arbitrary metrics in psychology. *American Psychologist*, *61*(1), 27.

Bond, T. G., & Fox, C. M. (2013). *Applying the Rasch model: Fundamental measurement in the human sciences*. Psychology Press.

Boone, W. J. (2016). Rasch analysis for instrument development: Why, when, and how? *CBE—Life Sciences Education*, *15*(4), rm4.

Boone, W. J., & Scantlebury, K. (2006). The role of Rasch analysis when conducting science education research utilizing multiple-choice tests. *Science Education*, *90*(2), 253–269.

Boone, W. J., Staver, J. R., & Yale, M. S. (2013). *Rasch analysis in the human sciences*. Springer.

Boone, W. J., Staver, J. R., Yale, M. S., Boone, W. J., Staver, J. R., & Yale, M. S. (2014). Wright maps: First steps. *Rasch Analysis in the Human Sciences*, 111–136.

Bornstein, M. H., Jager, J., & Putnick, D. L. (2013). Sampling in developmental science: Situations, shortcomings, solutions, and standards. *Developmental Review*, *33*(4), 357–370.

Caleon, I., & Subramaniam, R. (2010). Development and application of a three-tier diagnostic test to assess secondary students' understanding of waves. *International Journal of Science Education*, *32*(7), 939–961.

Cartrette, D. P., & Melroe-Lehrman, B. M. (2012). Describing changes in undergraduate students' preconceptions of research activities. *Research in Science Education*, *42*, 1073–1100.

Cetin-Dindar, A., & Geban, O. (2011). Development of a three-tier test to assess high school students' understanding of acids and bases. *Procedia-Social and Behavioral Sciences*, *15*, 600–604.

Chan, S. W., Ismail, Z., & Sumintono, B. (2014). A Rasch model analysis on secondary students' statistical reasoning ability in descriptive statistics. *Procedia-Social and Behavioral Sciences*, *129*, 133–139.

Chi, M. T. H. (2005). Commonsense conceptions of emergent processes: Why some misconceptions are robust. *The Journal of the Learning Sciences*, *14*(2), 161–199.

Colton, D., & Covert, R. W. (2007). *Designing and constructing instruments for social research and evaluation*. John Wiley & Sons.

Coştu, B. (2008). Learning science through the PDEODE teaching strategy: Helping students make sense of everyday situations. *Eurasia Journal of Mathematics, Science and Technology Education*, *4*(1), 3–9.

Didiş, N., Eryılmaz, A., & Erkoç, Ş. (2014). Investigating students' mental models about the quantization of light, energy, and angular momentum. *Physical Review Special Topics-Physics Education Research*, *10*(2), 20127.

Eagly, A. H., Makhijani, M. G., & Klonsky, B. G. (1992). Gender and the evaluation of leaders: A meta-analysis. *Psychological Bulletin*, *111*(1), 3.

Elo, S., Kääriäinen, M., Kanste, O., Pölkki, T., Utriainen, K., & Kyngäs, H. (2014). Qualitative content analysis: A focus on trustworthiness. *SAGE Open*, *4*(1), 2158244014522633.

Eryilmaz, A. (2010). Development and application of three-tier heat and temperature test: Sample of bachelor and graduate students. *Eurasian Journal of Educational Research*, *40*(1), 53–76.

Eshetu, F., & Alemu, M. (2018). Students conception of voltage and resistance concepts after conventional instruction. *EURASIA Journal of Mathematics, Science and Technology Education*, *14*(7), 3021–3033.

Fosnot, C. T., & Perry, R. S. (1996). Constructivism: A psychological theory of learning. *Constructivism: Theory, Perspectives, and Practice*, *2*(1), 8–33.

Gurcay, D., & Gulbas, E. (2015). Development of three-tier heat, temperature and internal energy diagnostic test. *Research in Science & Technological Education*, *33*(2), 197–217.

Gurel, D. K., Eryilmaz, A., & McDermott, L. C. (2015). A review and comparison of diagnostic instruments to identify students' misconceptions in science. *Eurasia Journal of Mathematics, Science and Technology Education*, *11*(5), 989–1008. https://doi.org/10.12973/eurasia.2015.1369a

Ivanjek, L., Morris, L., Schubatzky, T., Hopf, M., Burde, J.-P., Haagen-Schützenhöfer, C., Dopatka, L., Spatz, V., & Wilhelm, T. (2021). Development of a two-tier instrument on simple electric circuits. *Physical Review Physics Education Research*, *17*(2), 20123.

Kaltakci-Gurel, D., Eryilmaz, A., & McDermott, L. C. (2017). Development and application of a four-tier test to assess pre-service physics teachers' misconceptions about geometrical optics. *Research in Science and Technological Education*, *35*(2), 238–260. https://doi.org/10.1080/02635143.2017.1310094

Kaltakci, D., & Didis, N. (2007). Identification of pre-service physics teachers' misconceptions on gravity concept: A study with a 3-tier misconception test. *Sixth International Conference on Balkan Physical Union(AIP Conference Proceedings Volume 899)*, *899*, 499–500.

Keeley, P. (2012). Misunderstanding misconceptions. *Science Scope*, *35*(8), 12–15.

Kimberlin, C. L., & Winterstein, A. G. (2008). Validity and reliability of measurement instruments used in research. *American Journal of Health-System Pharmacy*, *65*(23), 2276–2284.

Kızılcık, H., & Güneş, B. (2011). Developing three-tire misconception test about regular circular motion. *HACETTEPE UNIVERSITESI EGITIM FAKULTESI DERGISI-HACETTEPE UNIVERSITY JOURNAL OF EDUCATION*.

Korkmaz, C., Ay, Ö., Ersoysal, Y., Köroğlu, M. A., & Erdem, C. (2019). Heavy metal levels in muscle tissues of some fish species caught from north-east Mediterranean: Evaluation of their effects on human health. *Journal of Food Composition and Analysis*, *81*, 1–9.

Kulgemeyer, C., & Wittwer, J. (2023). Misconceptions in physics explainer videos and the illusion of understanding: An experimental study. *International Journal of Science and Mathematics Education*, *21*(2), 417–437.

Liu, G., & Fang, N. (2016). Student misconceptions about force and acceleration in physics and engineering mechanics education. *International Journal of Engineering Education*, *32*(1), 19–29.

Moss-Racusin, C. A., Dovidio, J. F., Brescoll, V. L., Graham, M. J., & Handelsman, J. (2012). Science faculty's subtle gender biases favor male students. *Proceedings of the National*

Academy of Sciences, *109*(41), 16474–16479.

Myers, N. D., Wolfe, E. W., Feltz, D. L., & Penfield, R. D. (2006). Identifying differential item functioning of rating scale items with the Rasch model: An introduction and an application. *Measurement in Physical Education and Exercise Science*, *10*(4), 215–240.

Neumann, I., Neumann, K., & Nehm, R. (2011). Evaluating instrument quality in science education: Rasch-based analyses of a nature of science test. *International Journal of Science Education*, *33*(10), 1373–1405.

Özdemir, G., & Clark, D. B. (2007). An overview of conceptual change theories. *Eurasia Journal of Mathematics, Science and Technology Education*, *3*(4), 351–361.

Peşman, H., & Eryılmaz, A. (2010). Development of a three-tier test to assess misconceptions about simple electric circuits. *The Journal of Educational Research*, *103*(3), 208–222.

Petersen, L., Minkkinen, P., & Esbensen, K. H. (2005). Representative sampling for reliable data analysis: theory of sampling. *Chemometrics and Intelligent Laboratory Systems*, *77*(1–2), 261–277.

Piaget, J. (1976). *Piaget's theory*. Springer.

Planinic, M., Boone, W. J., Susac, A., & Ivanjek, L. (2019). Rasch analysis in physics education research: Why measurement matters. *Physical Review Physics Education Research*, *15*(2), 20111.

Potvin, P., & Cyr, G. (2017). Toward a durable prevalence of scientific conceptions: Tracking the effects of two interfering misconceptions about buoyancy from preschoolers to science teachers. *Journal of Research in Science Teaching*, *54*(9), 1121–1142.

Putnick, D. L., & Bornstein, M. H. (2016). Measurement invariance conventions and reporting: The state of the art and future directions for psychological research. *Developmental Review*, *41*, 71–90.

Quaigrain, K., & Arhin, A. K. (2017). Using reliability and item analysis to evaluate a teacher-developed test in educational measurement and evaluation. *Cogent Education*, *4*(1), 1301013.

Resbiantoro, G., & Setiani, R. (2022). A review of misconception in physics: the diagnosis, causes, and remediation. *Journal of Turkish Science Education*, *19*(2).

Rompayom, P., Tambunchong, C., Wongyounoi, S., & Dechsri, P. (2011). Using Open-Ended Questions to Diagnose Students' Understanding of Inter-and Intramolecular Forces. *Online Submission*.

Roth, K. J. (2013). Developing meaningful conceptual understanding in science. In *Dimensions of thinking and cognitive instruction* (pp. 139–175). Routledge.

Saidi, S. S., & Siew, N. M. (2019). Reliability and Validity Analysis of Statistical Reasoning Test Survey Instrument Using the Rasch Measurement Model. *International Electronic Journal of Mathematics Education, 14*(3), 535–546.

Samsudin, A., Afif, N. F., Nugraha, M. G., Suhandi, A., Fratiwi, N. J., Aminudin, A. H., Adimayuda, R., Linuwih, S., & Costu, B. (2021). Reconstructing students' misconceptions on work and energy through the PDEODE* E tasks with think-pair-share. *Journal of Turkish Science Education*, *18*(1), 118–144.

Samsudin, A., Zulfikar, A., Saepuzaman, D., Suhandi, A., Aminudin, A. H., Supriyadi, S., & Coştu, B. (2024). Correcting grade 11 students' misconceptions of the concept of force through the conceptual change model (CCM) with PDEODE* E tasks. *Journal of Turkish Science Education*, *21*(2), 212–231.

Segars, A. H. (1997). Assessing the unidimensionality of measurement: A paradigm and illustration within the context of information systems research. *Omega*, *25*(1), 107–121.

Sesli, E., & Kara, Y. (2012). Development and application of a two-tier multiple-choice diagnostic test for high school students' understanding of cell division and reproduction. *Journal of Biological Education*, *46*(4), 214–225.

Soeharto, S., CsapÃ, B., Sarimanah, E., Dewi, F. I., & Sabri, T. (2019). A review of studentsâ€™ common misconceptions in science and their diagnostic assessment tools. *Jurnal Pendidikan IPA Indonesia*, *8*(2), 247–266.

Stefanidou, C., & Skordoulis, C. (2017). Primary student teachers' understanding of basic ideas of nature of science: Laws, theories and models. *Journal of Studies in Education*, *7*(1), 127–153.

Sumintono, B., & Widhiarso, W. (2014a). *Aplikasi model Rasch untuk penelitian ilmu-ilmu sosial (edisi revisi)*. Trim Komunikata Publishing House.

Sumintono, B., & Widhiarso, W. (2014b). Aplikasi Model Rasch untuk Penelitian Ilmu-Ilmu Sosial [Application of the Rasch Model for Social Sciences Research]. *Bandung, Indonesia: Trimkom Publishing House*.

Sun, Y.-C. (2009). Using a two-tier test in examining Taiwan graduate students' perspectives on paraphrasing strategies. *Asia Pacific Education Review, 10,* 399–408.

Taslidere, E. (2016). Development and use of a three-tier diagnostic test to assess high school students' misconceptions about the photoelectric effect. *Research in Science & Technological Education*, *34*(2), 164–186.

TOPALSAN, A. K., & Bayram, H. (2019). Identifying Prospective Primary School Teachers' Ontologically Categorized Misconceptions on the Topic of" Force and Motion". *Journal* *of Turkish Science Education*, *16*(1), 85–109.

Treagust, D. F. (1988). Development and use of diagnostic tests to evaluate students' misconceptions in science. *International Journal of Science Education*, *10*(2), 159–169.

Vosniadou, S. (2007). The cognitive-situative divide and the problem of conceptual change. *Educational Psychologist*, *42*(1), 55–66.

Wang, N. (2003). Use of the Rasch IRT model in standard setting: An item-mapping method. *Journal of Educational Measurement*, *40*(3), 231–253.

Wright, B. D. (1977). Solving measurement problems with the Rasch model. *Journal of Educational Measurement*, 97–116.

BIOGRAPHIES OF AUTHORS

Mohd Zaidi Bin Amiruddin . He is a student in the Science Education Programme, Faculty of Mathematics and Natural Science Education, Universitas Pendidikan Indonesia, Bandung, Indonesia. His research interests are conceptual change, model mental, and misconception. He can be contacted at email: mohdzaidi@upi.edu.

Achmad Samsudin is a lecturer in Physics Education Programme, Faculty of Mathematics and Natural Science Education, Universitas Pendidikan Indonesia, Bandung, Indonesia. His current research interests are physics education, science education, and cognitive psychology such as identification of conceptions, misconceptions, conceptual change, and development of models, instruments, and media. He can be contacted at email: achmadsamsudin@upi.edu.

Andi Suhandi is a lecturer at Science Education Programme, Universitas Pendidikan Indonesia, Bandung, Indonesia. He is also a professor at the Universitas Pendidikan Indonesia in the Field of Physics. His current research interests are physics, materials physics, physics education, science education, and teaching and learning such as conception, misconceptions, and conceptual change text. He can be contacted at email: andi_sh@upi.edu.

Bayram Costu is a lecturer at Department of Science Education, Yildiz Technical University, Istanbul, Turkey. He is also a professor at the Yildiz Technical University. His current research interests are teaching and learning, science education, conception, misconceptions, and conceptual change. He can be contacted at email: bayramcostu@gmail.com.

Adam Hadiana Aminudin is a lecturer and researcher in Physics, Faculty of Mathematics and Natural Sciences, Universitas Kebangsaan Republik Indonesia, Bandung, Indonesia. The focus of the research studied includes physics, teaching, development of renewable technology-based media, conceptions, misconceptions, and conceptual changes. He can be contacted at email: adamhadianaaminudin@mipa.ukri.ac.id

BIOGRAPHIES OF AUTHORS

Mohd Zaidi Bin Amiruddin . He is a student in the Science Education Programme, Faculty of Mathematics and Natural Science Education, Universitas Pendidikan Indonesia, Bandung, Indonesia. His research interests are conceptual change, model mental, and misconception. He can be contacted at email: mohdzaidi@upi.edu.

Achmad Samsudin is a lecturer in Physics Education Programme, Faculty of Mathematics and Natural Science Education, Universitas Pendidikan Indonesia, Bandung, Indonesia. His current research interests are physics education, science education, and cognitive psychology such as identification of conceptions, misconceptions, conceptual change, and development of models, instruments, and media. He can be contacted at email: achmadsamsudin@upi.edu.

Andi Suhandi is a lecturer at Science Education Programme, Universitas Pendidikan Indonesia, Bandung, Indonesia. He is also a professor at the Universitas Pendidikan Indonesia in the Field of Physics. His current research interests are physics, materials physics, physics education, science education, and teaching and learning such as conception, misconceptions, and conceptual change text. He can be contacted at email: andi_sh@upi.edu.

Bayram Costu is a lecturer at Department of Science Education, Yildiz Technical University, Istanbul, Turkey. He is also a professor at the Yildiz Technical University. His current research interests are teaching and learning, science education, conception, misconceptions, and conceptual change. He can be contacted at email: bayramcostu@gmail.com.

Adam Hadiana Aminudin is a lecturer and researcher in Physics, Faculty of Mathematics and Natural Sciences, Universitas Kebangsaan Republik Indonesia, Bandung, Indonesia. The focus of the research studied includes physics, teaching, development of renewable technology-based media, conceptions, misconceptions, and conceptual changes. He can be contacted at email: adamhadianaaminudin@mipa.ukri.ac.id

Universitas Pendidikan Indonesia

MOHD ZAIDI <mohdzaidi@upi.edu>

Your Submission

6 messages

Social Sciences & Humanities Open <em@editorialmanager.com> Tue, Mar 11, 2025 at 5:46 PM
Reply-To: Social Sciences & Humanities Open <support@elsevier.com>
To: Mohd Zaidi Bin Amiruddin <mohdzaidi@upi.edu>

Ref.: Ms. No. **SSHO-D-24-03178**
Revealing High School Students' Misconceptions Using 3-Tier Diagnostic Open-Ended Test (3-DOET) about Linear Motion: A Rasch Measurement Approach
Social Sciences & Humanities Open

Dear Mr Bin Amiruddin,

Reviewers have now commented on your paper. You will see that they are advising that you revise your manuscript. While one reviewer asked to reject your manuscript, I am providing you with an opportunity to revise your manuscript based on their comments. If you are prepared to undertake the work required, I would be pleased to reconsider my decision.

For your guidance, editor's and reviewers' comments are appended below.

If you decide to revise the work, please submit a list of changes or a rebuttal against each point which is being raised when you submit the revised manuscript.

Your revision is due by **Apr 01, 2025**.

To submit a revision, go to https://www.editorialmanager.com/ssho/ and log in as an Author. You will see a menu item call Submission Needing Revision. You will find your submission record there.

Research Elements (optional)
This journal encourages you to share research objects - including your raw data, methods, protocols, software, hardware and more – which support your original research article in a Research Elements journal. Research Elements are open access, multidisciplinary, peer-reviewed journals which make the objects associated with your research more discoverable, trustworthy and promote replicability and reproducibility. As open access journals, there may be an Article Publishing Charge if your paper is accepted for publication. Find out more about the Research Elements journals at https://www.elsevier.com/authors/tools-and-resources/research-elements-journals?dgcid=ec_em_research_elements_email.

Yours sincerely

Faranak Gholampour, Ph.D.
Scientific Handling Editor
Social Sciences & Humanities Open

Comments from the Editors and Reviewers:

Reviewer's Responses to Questions

Note: In order to effectively convey your recommendations for improvement to the author(s), and help editors make well-informed and efficient decisions, we ask you to answer the following specific questions about the manuscript and provide additional suggestions where appropriate.

1. Do the authors explain the reason for writing a review article in this field?

Please provide suggestions to the author(s) on how to better justify their reasons. Please number each suggestion so that the author(s) can more easily respond.

Reviewer #1: The background of this study has not shown the specific purpose of this study.
The direction of the new study can be seen in the research questions asked.
Please improve it with specifics and details for the background of this study.
Why do the research, the reasons underlying it, how can it be clarified.

Reviewer #2: The justification of the topic must be clear, especially why this study focus on 'linear motion', 'misconception, 'high school students'?

---

2. Does the review article provide a good overview of the development of the field while providing insights on its future development?

Please list the historical developments of likely future scenarios that the author(s) should add or emphasize more. Please number each suggestion so that the author(s) can more easily respond.

Reviewer #1: The article has shown the history of previous research, but the continuation and steps for conducting this research are not yet clear, so it is necessary to explore again how and what kind of research to do after knowing the development of the research.

Reviewer #2: The information about the Indonesian students' performance in topics is too brief.

---

3. Do authors adequately represent the most relevant and recent advances in the field?

Please provide suggestions to the author(s) on how to improve their reference list to include the relevant topics and cover both historical references and recent developments. Please number each suggestion so that the author(s) can more easily respond.

Reviewer #1: The author has used their reference list to cover relevant topics and includes historical references and current developments. However, it needs to be added and improved again with the latest references, because in the discussion there are still many articles that are more than 10 years old.

Reviewer #2: 1. Author did not present the current issues or problem regarding the topic of linear motion among the Indonesian high school students.
2. Author did not present the types of misconceptions for the topics.
3. Author did not present the theoretical framework of study.

---

4. Is the review reported in sufficient detail to allow for its replicability and/or reproducibility (e.g., search strategies disclosed, inclusion criteria and risk of bias assessment for individual studies stated, summary methods specified)?

Please provide suggestions to the author(s) on how to improve the replicability/reproducibility of their review. Please number each suggestion so that the author(s) can more easily respond.

Reviewer #1: The results written can be made more specific with the meaning or interpretation of the data that has been obtained with a specific explanation.

Reviewer #2: 1. 2.1 - what are the learning objectives of the topic involved in the curriculum?
2. 2.1 - Did author determine the content validity?
3. author did not explain clearly about Figure 2
4. 2.2 there is a big gap in terms of the number of students in grade 12 and grade 10 and 11?What is the reason?
5. What syllabus involved in the test?

---

5. Is the statistical summary method (e.g., meta-analysis, meta-regressions) and its reporting (e.g., P-values, 95%CIs, etc.) appropriate and well described?

Please clearly indicate if the review requires additional peer review by a statistician. Kindly provide suggestions to the author(s) on how to improve the statistical analyses, controls, sampling mechanism, or statistical reporting. Please number each suggestion so that the author(s) can more easily respond.

Reviewer #1: The use of statistical analysis is relevant, but the author needs to write the steps and specifically explain the meaning or interpretation of the data.

Reviewer #2: 1. Table 7, what types of misconceptions measured? The result were only about the levels of understanding. It did not represent the types of misconception of the topic.
2. The findings of study and the aims of study was not aligned as author did not report the types of misconception identified but only level of understanding.

---

6. Does the review structure, flow or writing need improving (e.g., the addition of subheadings, shortening of text, reorganization of sections, or moving details from one section to another, following PRISMA guidelines)?

Please provide suggestions to authors on how to improve the review structure and flow. Please number each suggestion so that the author(s) can more easily respond.

Reviewer #1: The author has compiled the article using Prisma steps, but it needs to be clarified in the data collection steps, data collection techniques and data analysis techniques.

Reviewer #2: 1. please provide a specific sub-heading to present the literature review about types of misconception of this topic. What are the gaps identified from the previous studies?
2. Did author determine the content validity?

---

7. Could the manuscript benefit from language editing?

Reviewer #1: Yes

Reviewer #2: Yes

Editor's comments:
Dear author,
To facilitate the peer review process, I kindly request the following revision:

Providing Ethical Approval or IRB Waiver Letter:
To proceed with the review process, please provide an ethical statement file issued by your institution detailing your study's ethical approval. The statement should include:
• The authors' names and the research title
• The full name of the approving ethics committee.
• The approval number.
• The date of the ethics approval.
Example format:
"This study was reviewed and approved by [full name of approving ethics committee] with the approval number: [approval number], dated [date of ethics approval] and signature."
I want to bring to your attention that the ethical approval letter needs to be obtained prior to conducting the study, not after the study was completed and post-submission.
Please include the ethical approval number and details regarding the informed consent obtained from participants in your manuscript.
If your institute does not routinely approve such research, please provide a formal waiver signed and dated by the Institutional Review Board (IRB).
Failure to demonstrate adherence to ethical protocols may result in rejection.

Paraphrasing to Reduce High Similarity and Plagiarism:
The manuscript contains sections that are highly similar to content from other sources. To address this:
• Revise and paraphrase any overlapping text while ensuring proper citation of the original sources.
• Re-check the manuscript using plagiarism detection software to verify that all issues are resolved.
This revision is essential to maintain originality and align with ethical publishing standards.

Additionally, please write the sources and references for all the figures.

Finally, please add DOI links to all references in your manuscript. This will enhance the accessibility and credibility of your citations.

The content of the topic only identified the five levels of students' understanding of the topic. It is not about revealing the 'misconception' as stated on the title. The content did not represent the title. Please explain this?

Bests Regards,
Faranak Gholampour, Ph.D.
Scientific Handling Editor

Reviewer #1: This field is optional. If you have any additional suggestions beyond those relevant to the questions above, please number and list them here.

Reviewer #2: This field is optional. If you have any additional suggestions beyond those relevant to the questions above, please number and list them here.